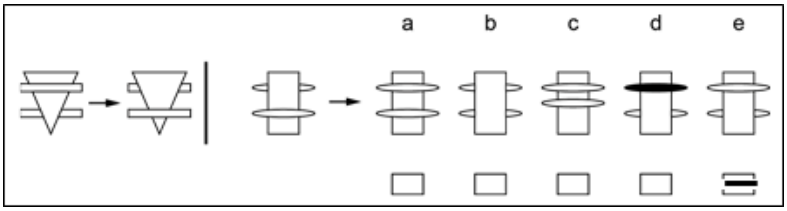
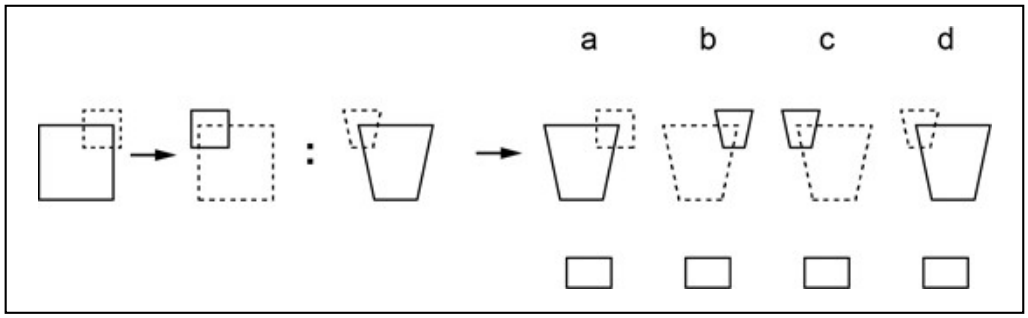
1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

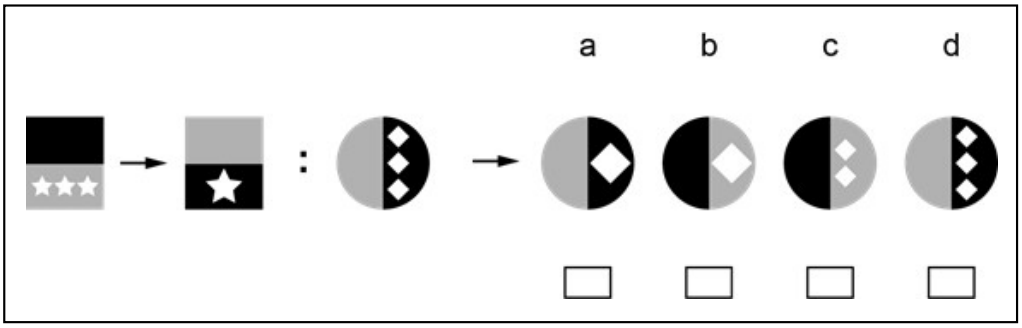
Example



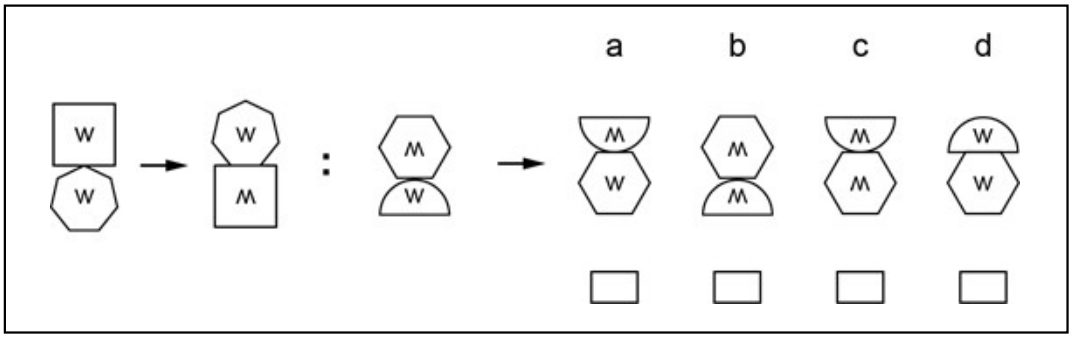
1



2



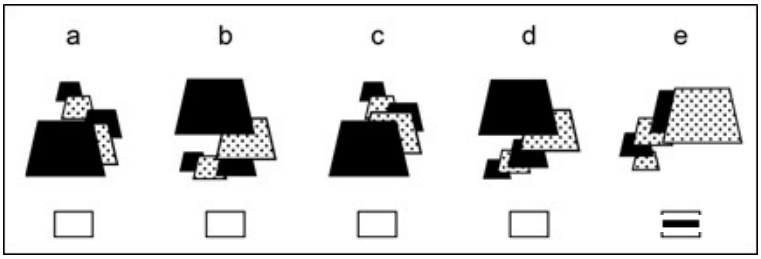
3



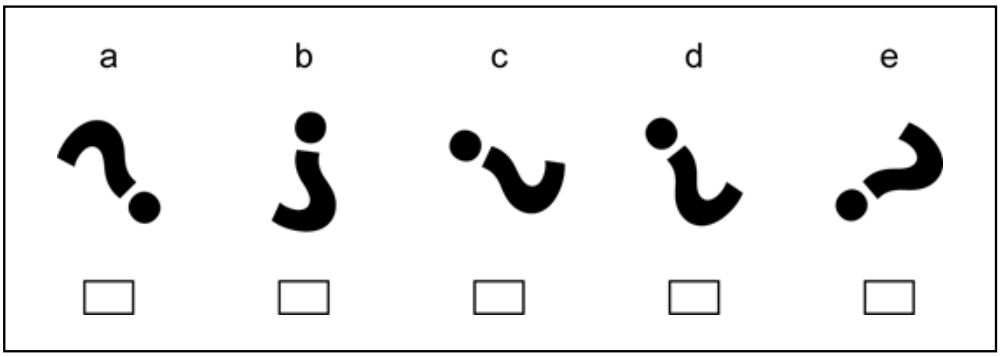
2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

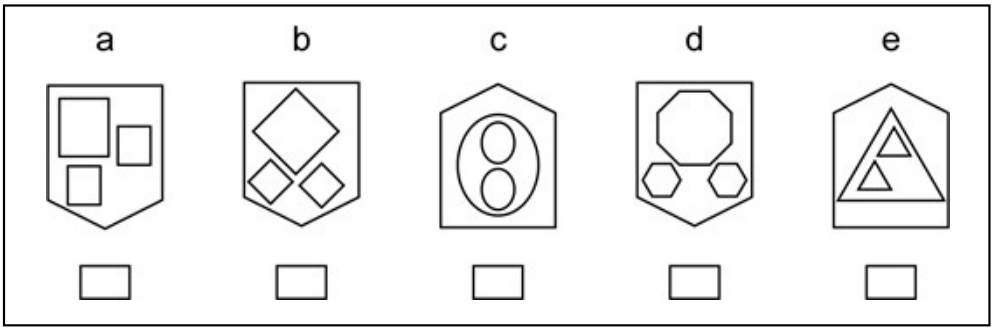
Example



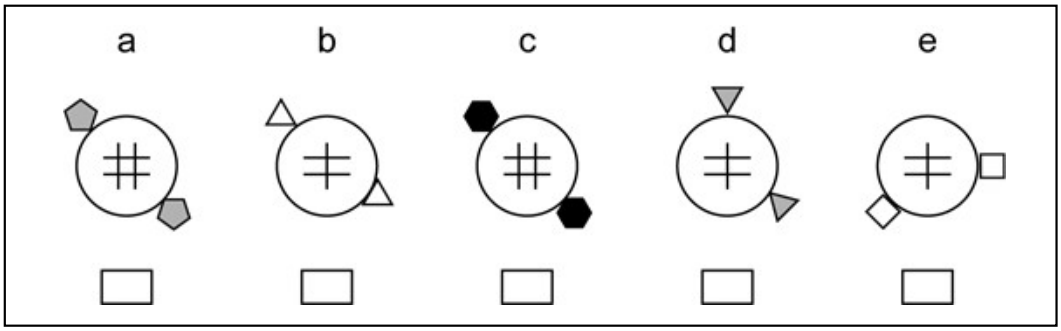
1



2



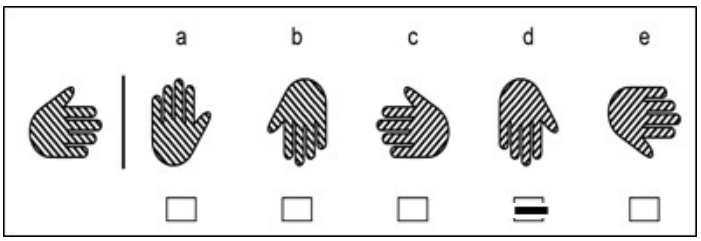
3



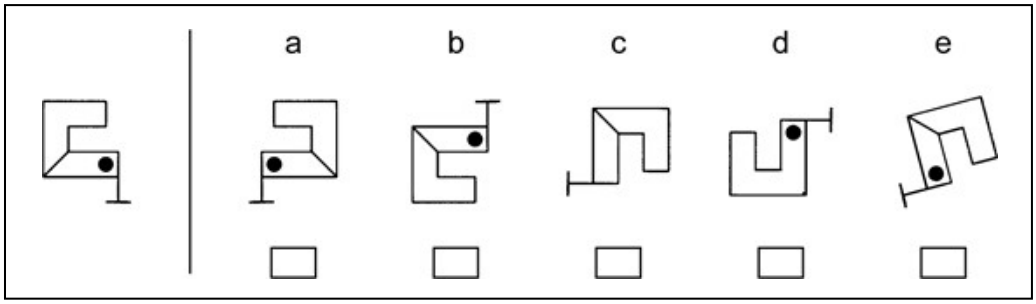
3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

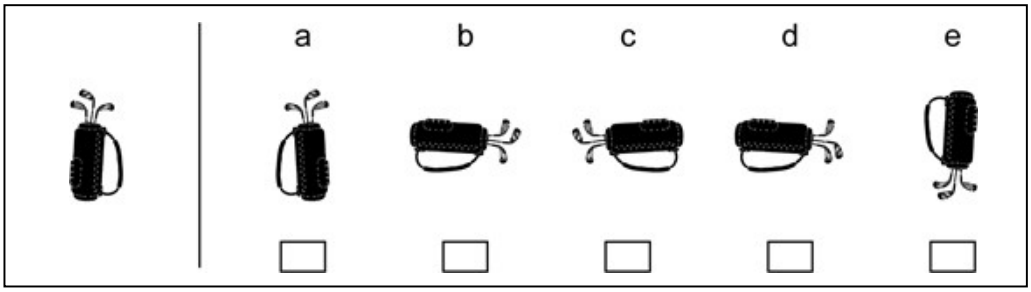
Example



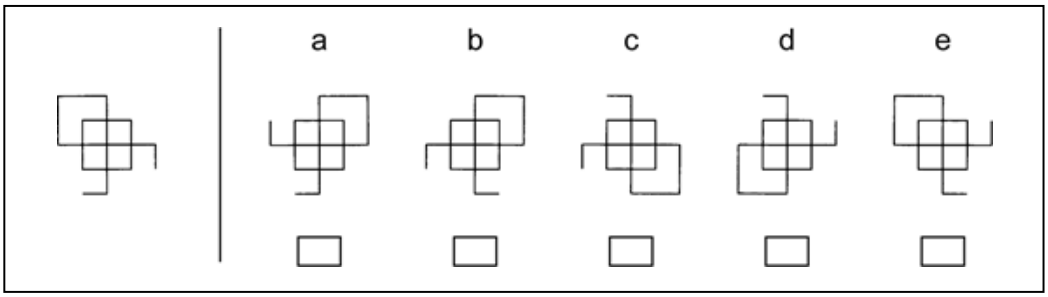
1



2



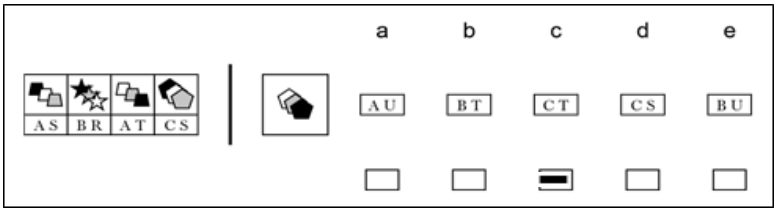
3



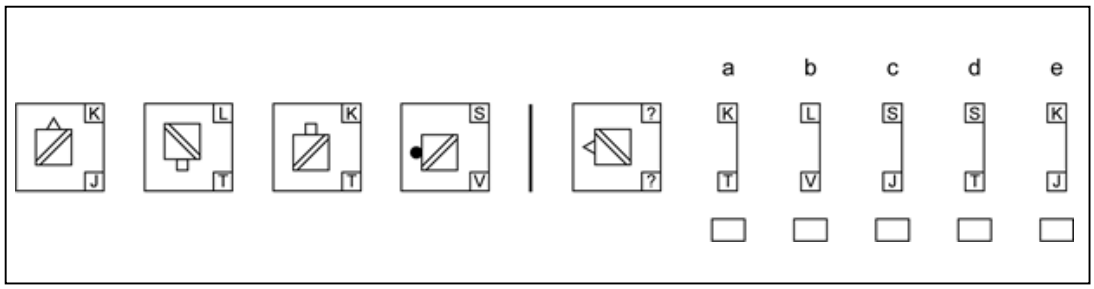
4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

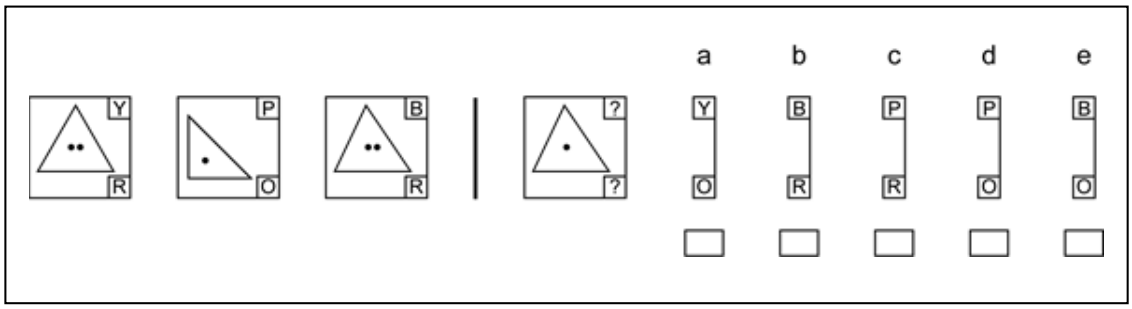
Example



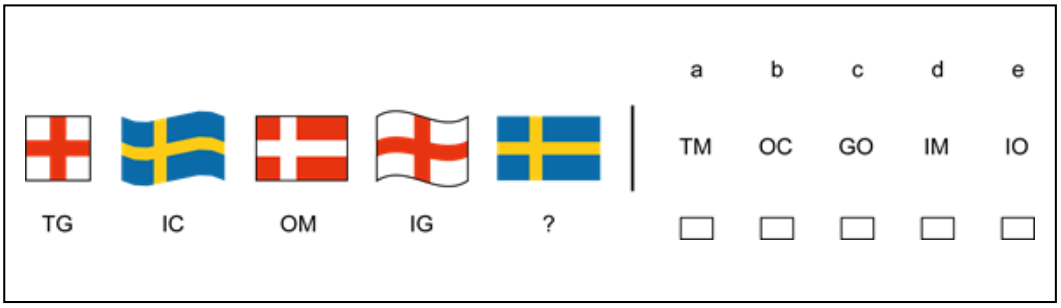
1



2



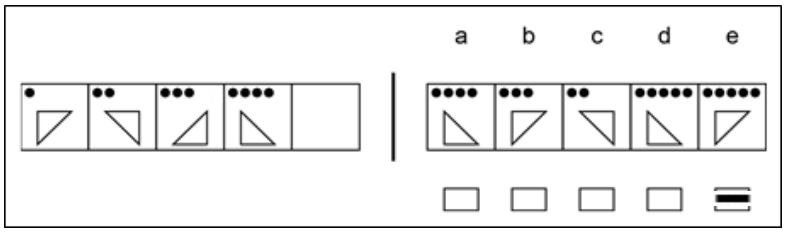
3



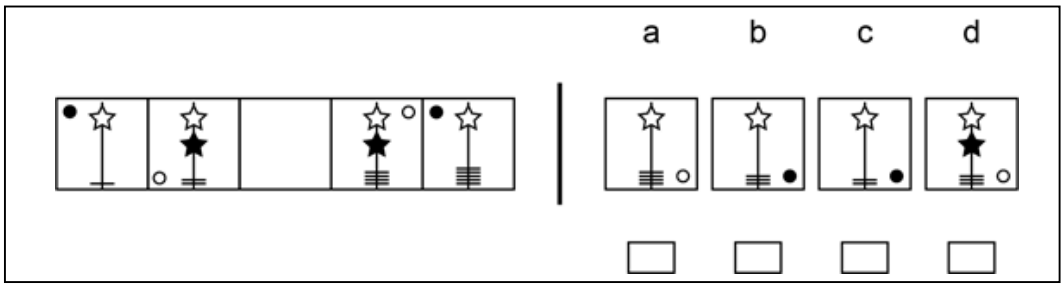
5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

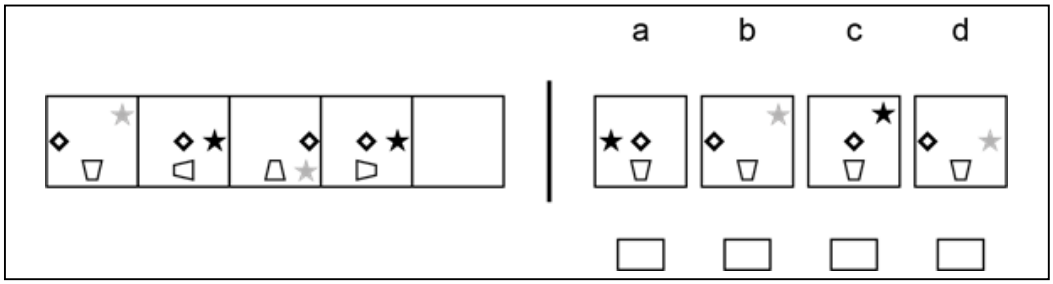
Example



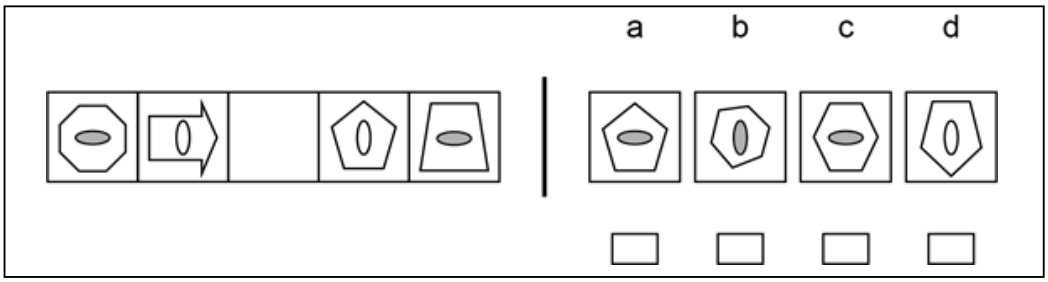
1



2



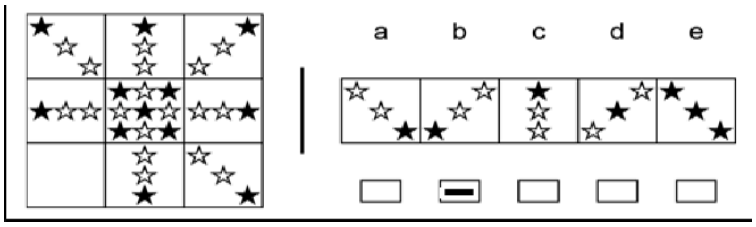
3



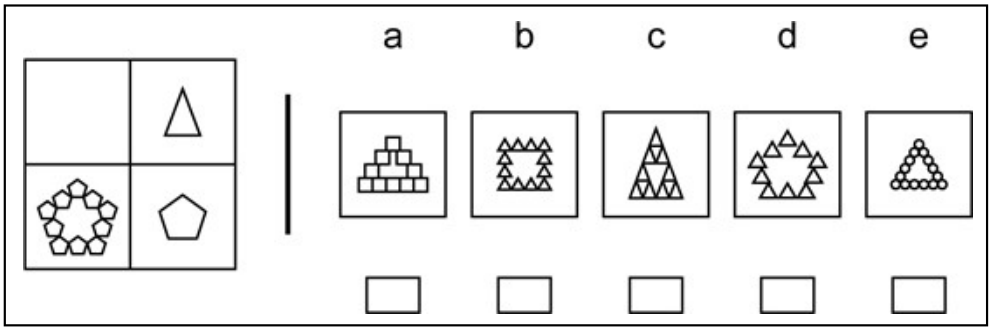
6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

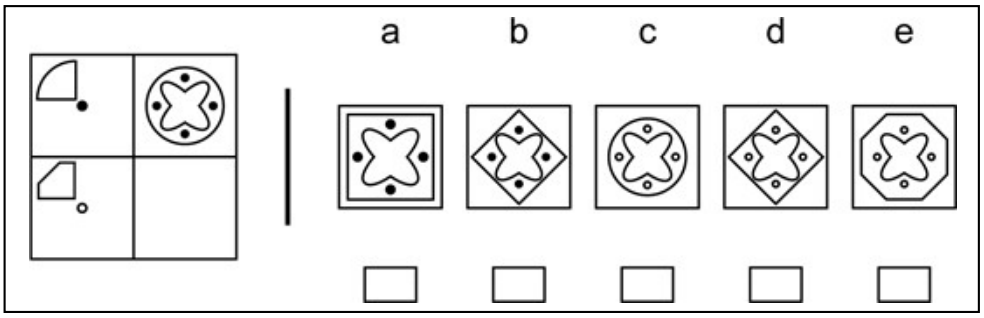
Example



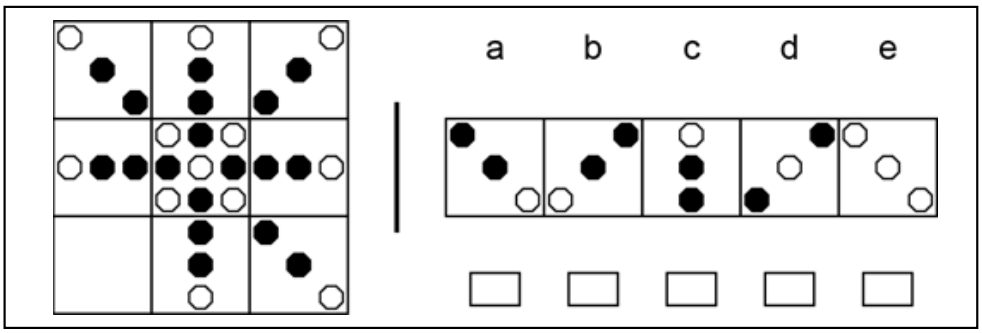
1



2



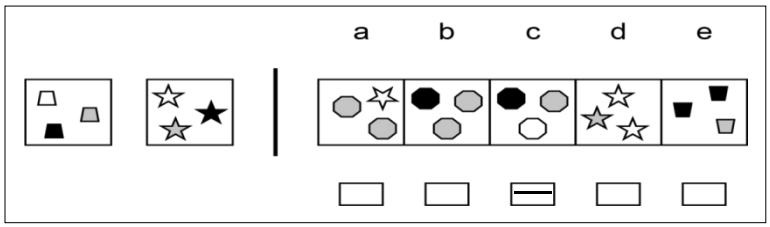
3



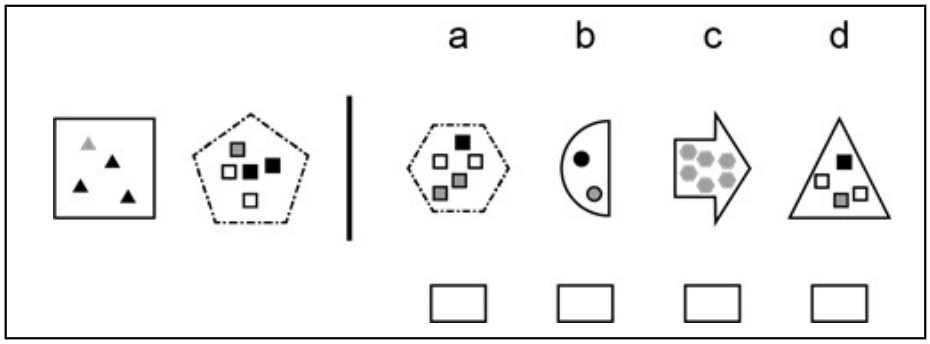
7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

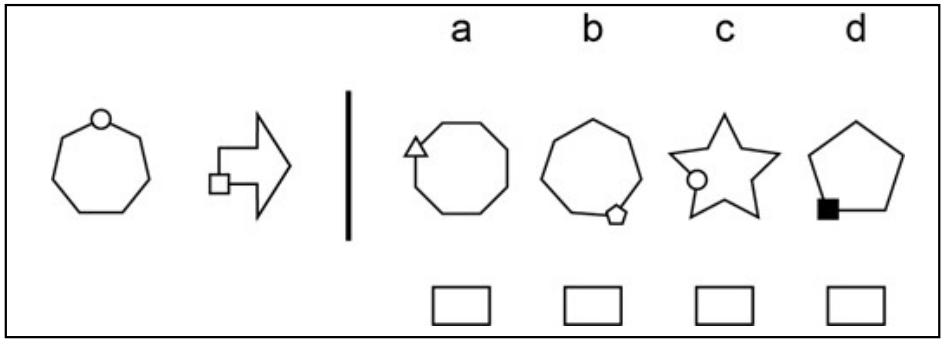
Example



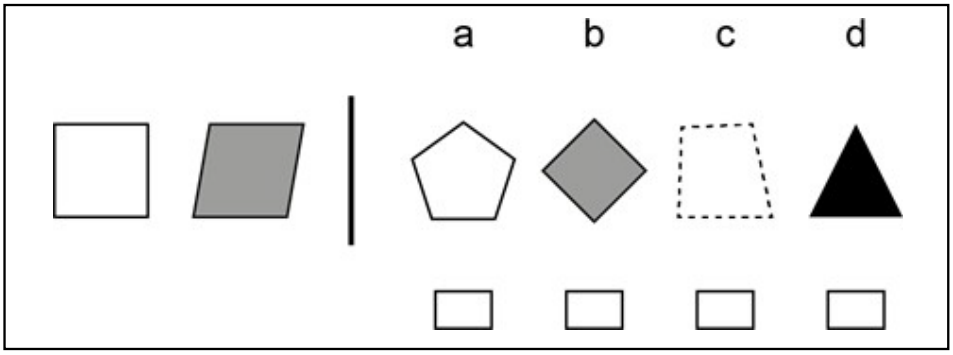
1



2



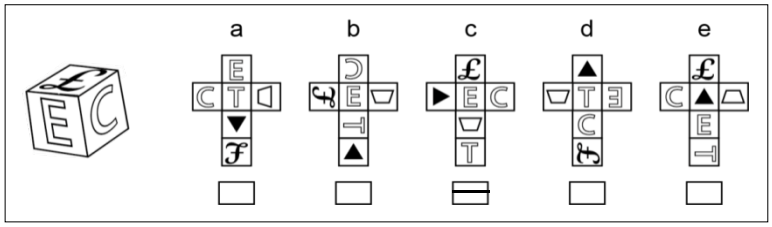
3



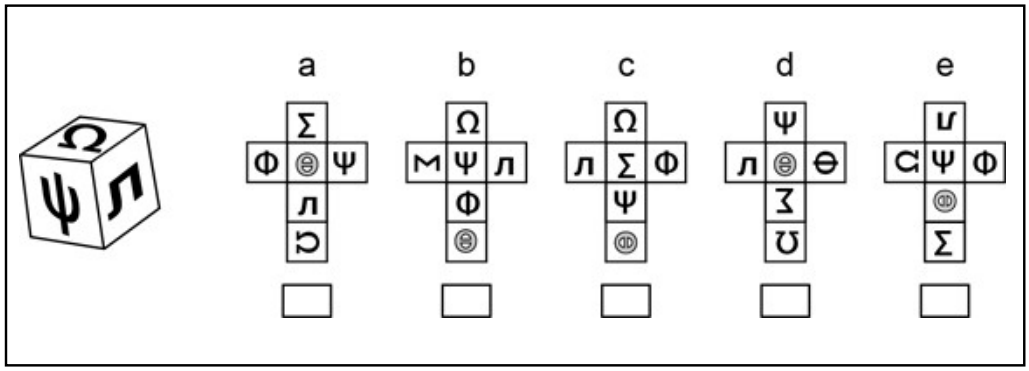
8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

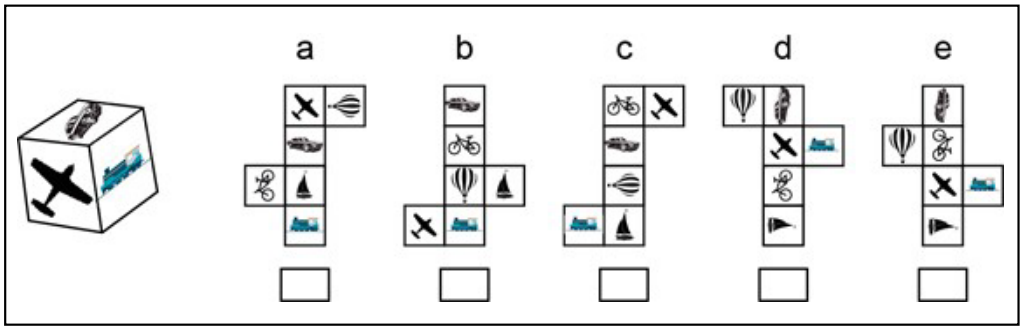
Example



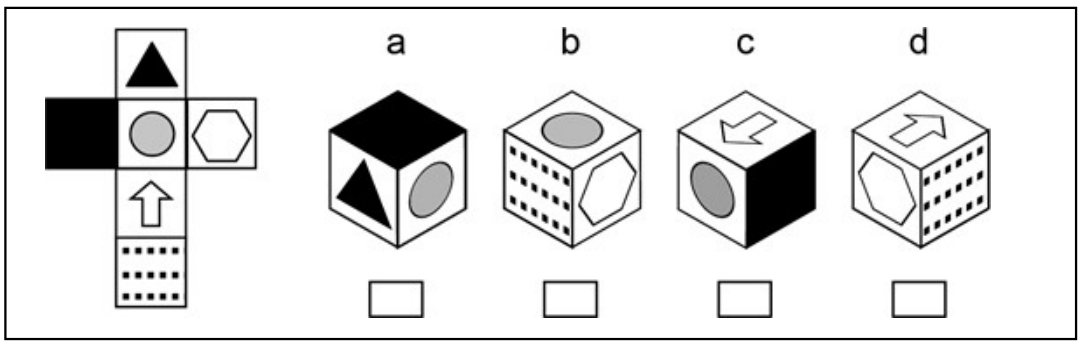
1



2



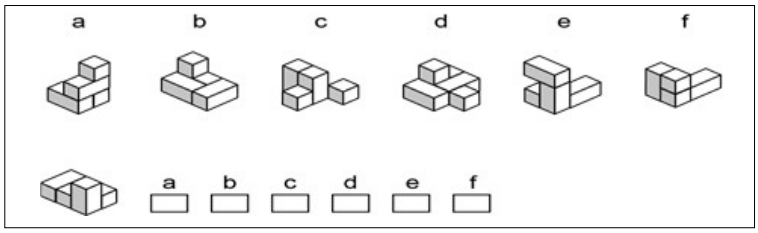
3



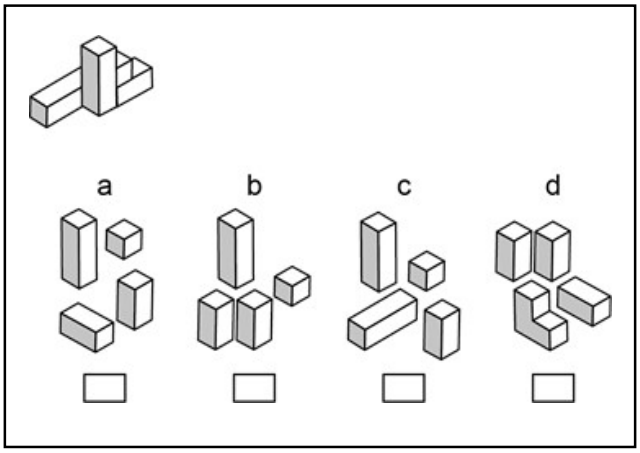
9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

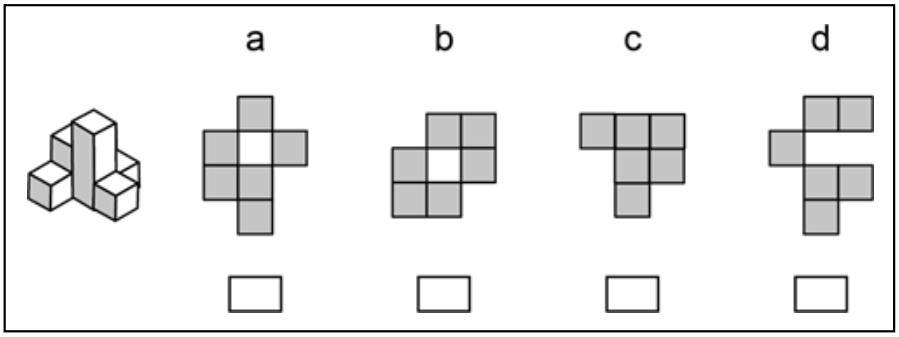
Example



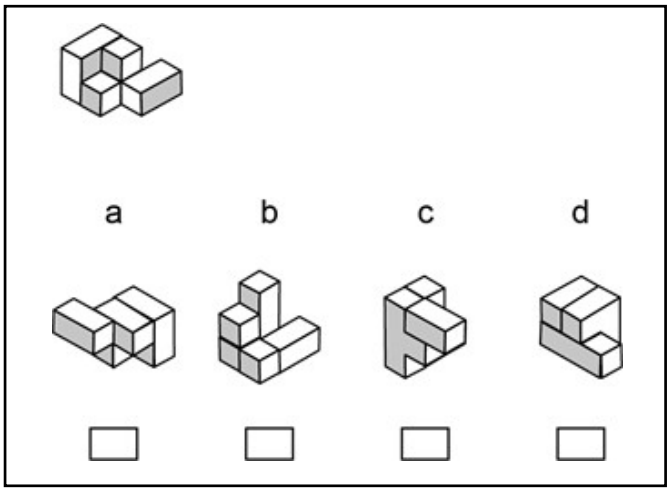
1



2



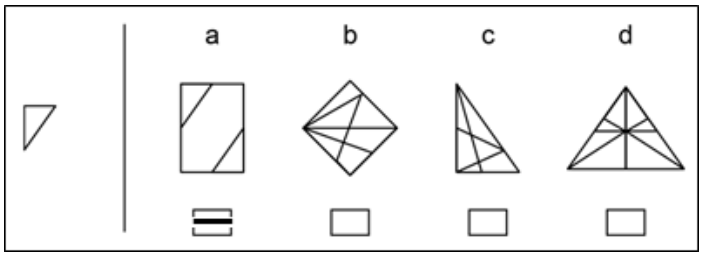
3



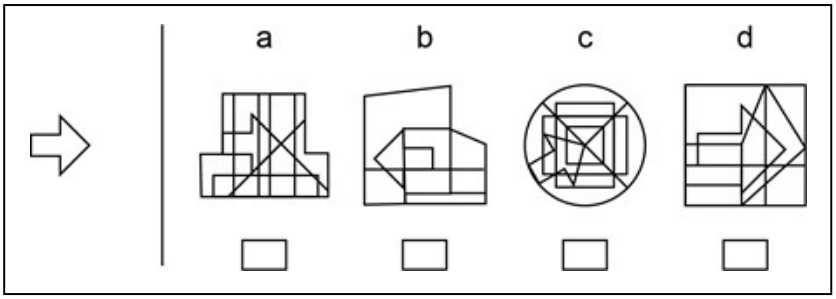
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

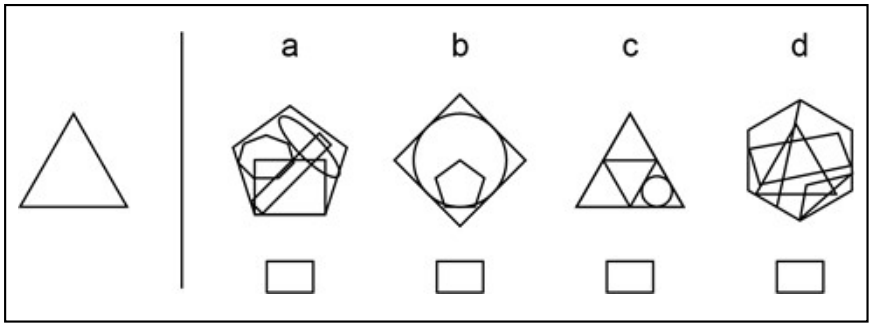
Example



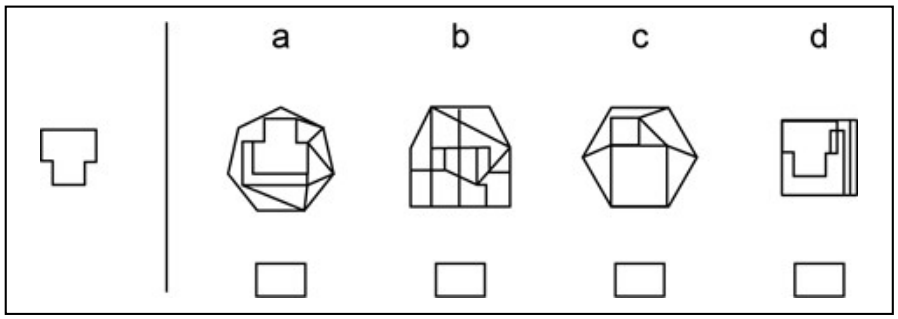
1



2



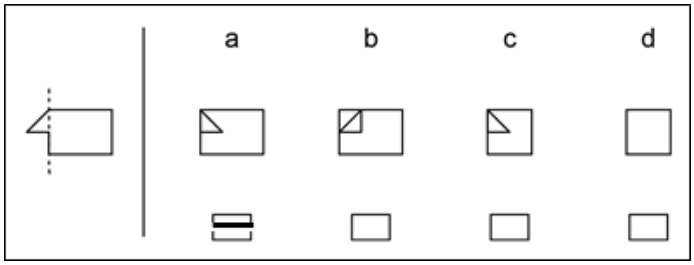
3



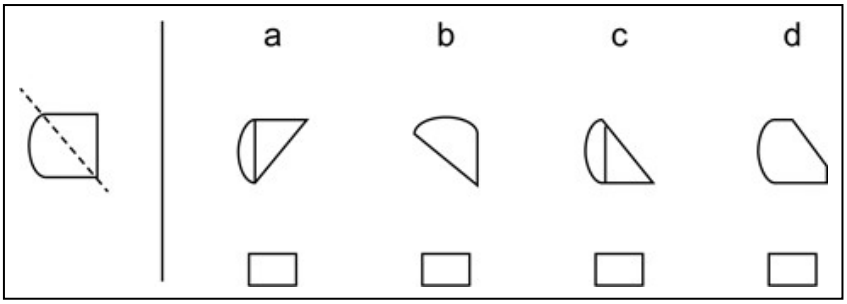
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

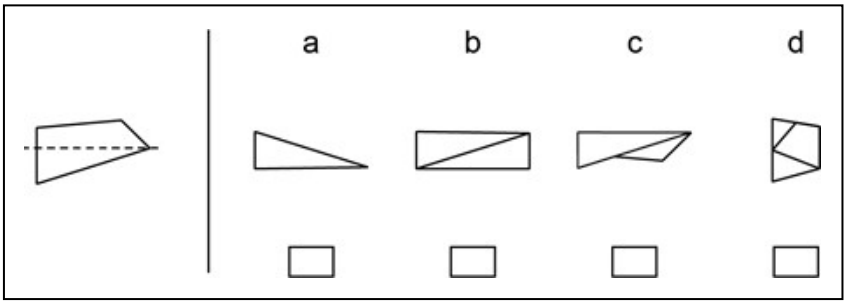
Example



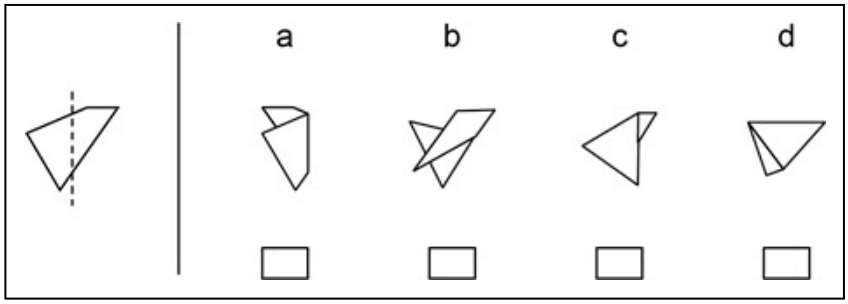
1



2



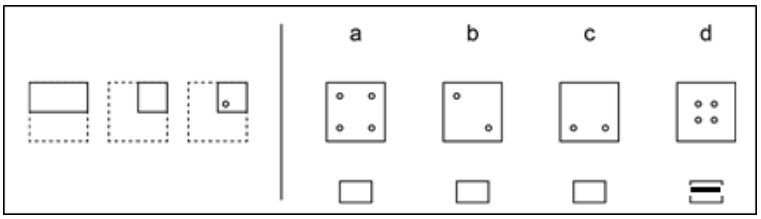
3



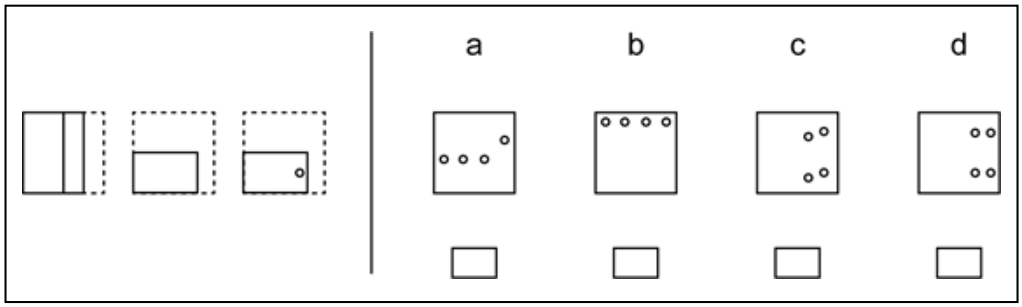
12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

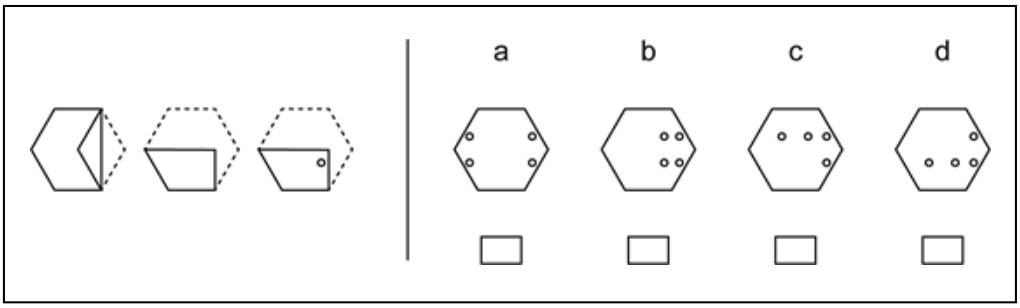
Example



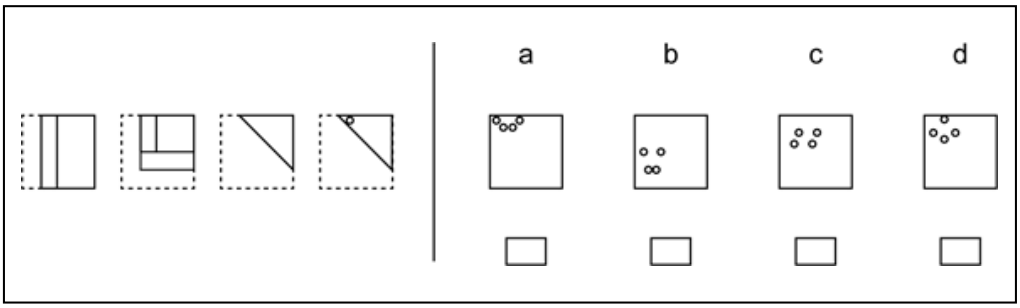
1



2



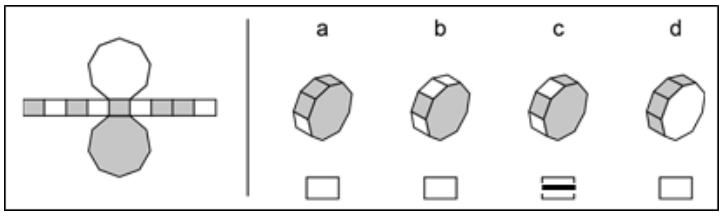
3



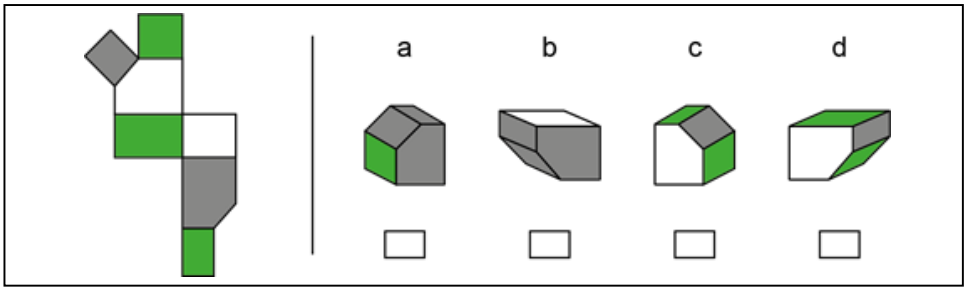
13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

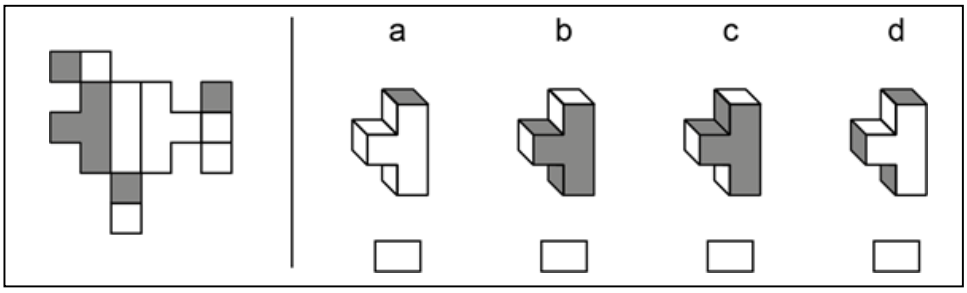
Example



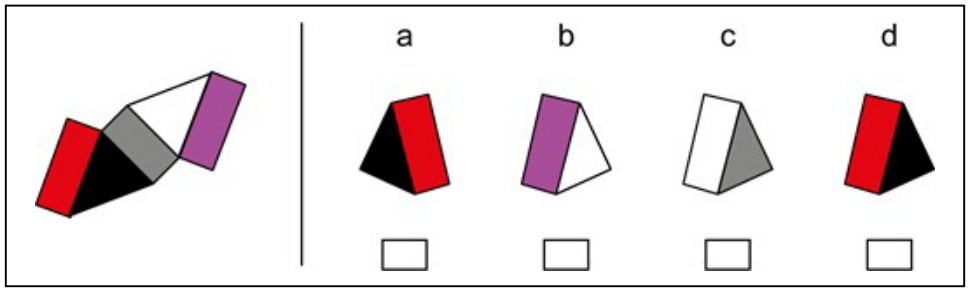
1



2



3

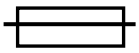


11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

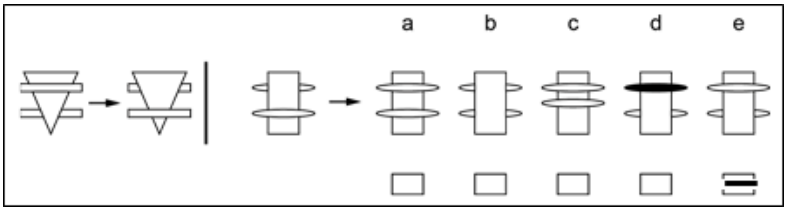
READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 25 minutes to complete the paper.
- Mark your answer like this: 
- This paper features a total of 39 questions from 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't answer a question, move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

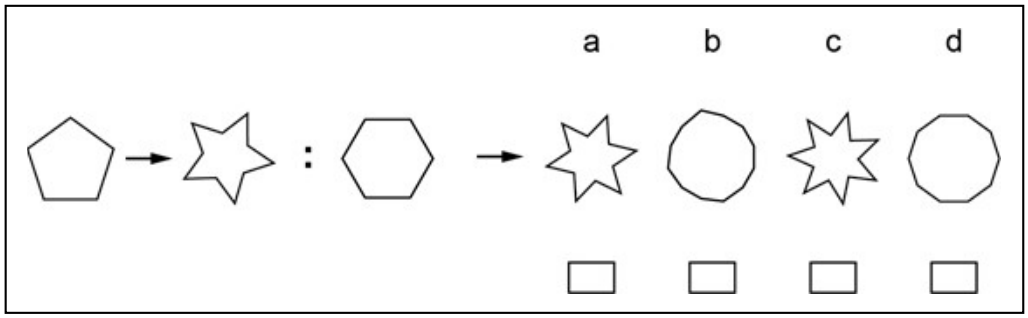
1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

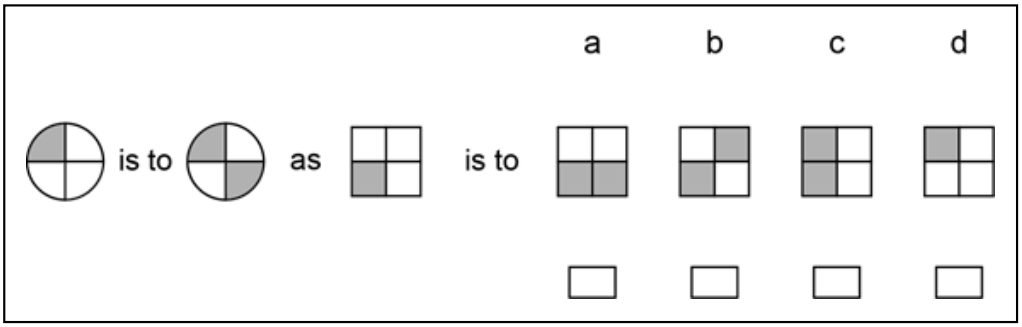
Example



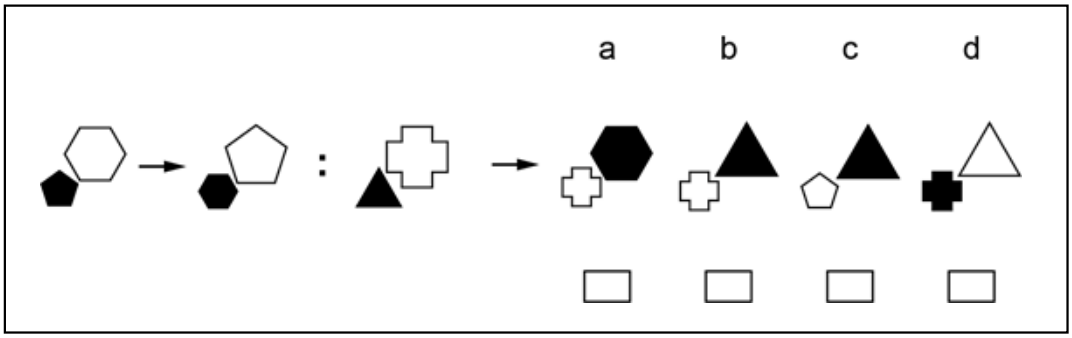
1



2



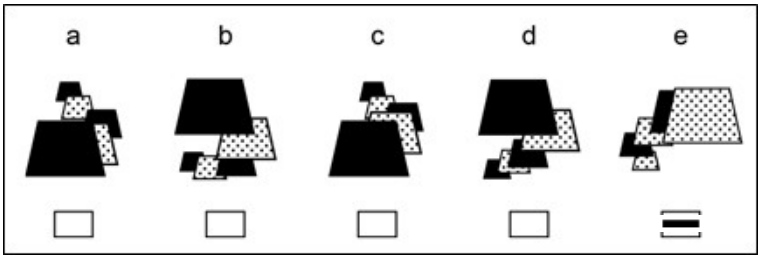
3



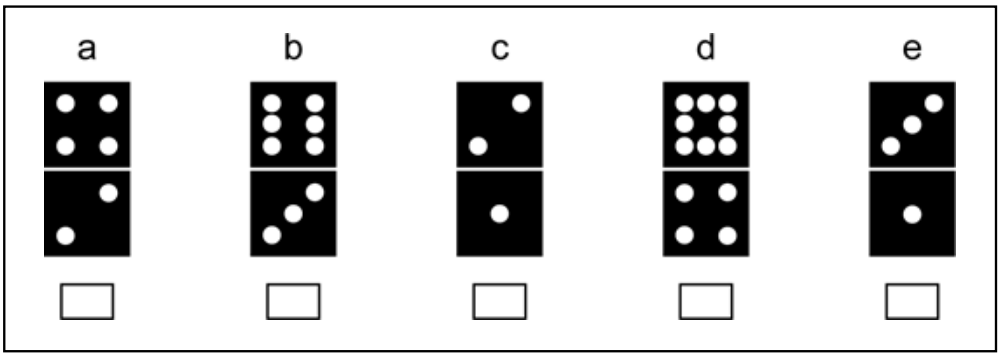
2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

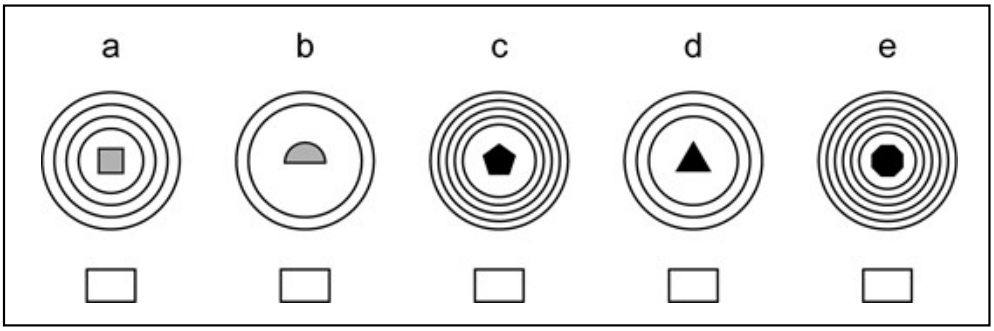
Example



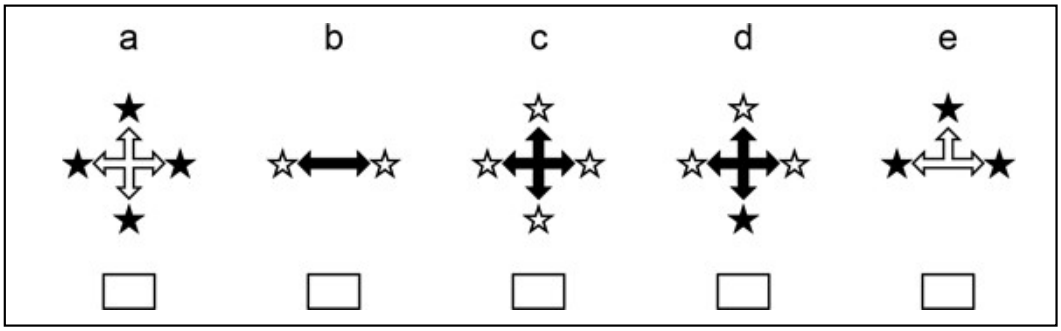
1



2



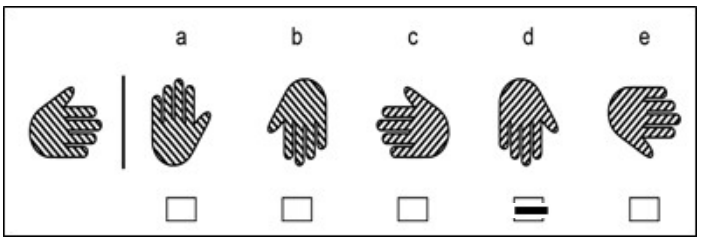
3



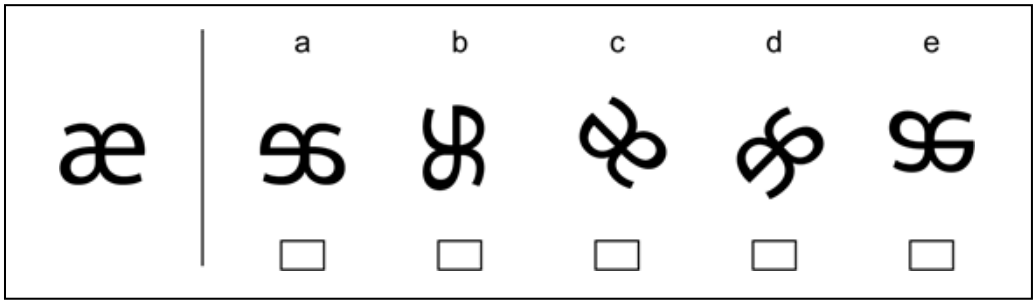
3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

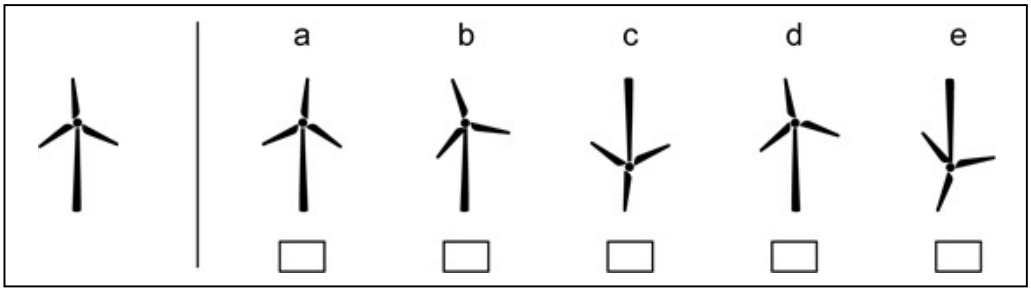
Example



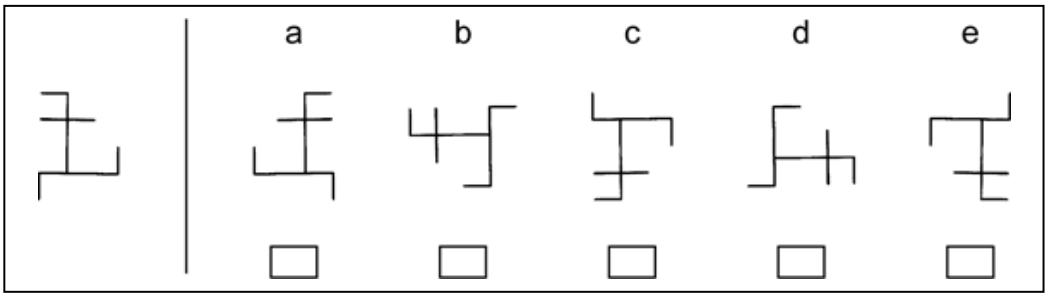
1



2



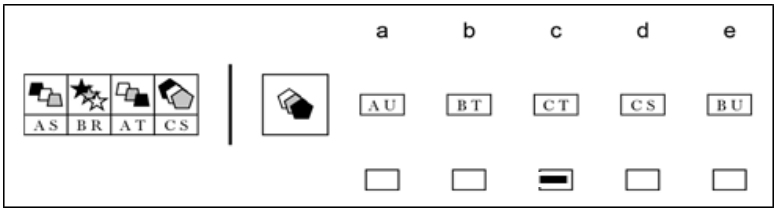
3



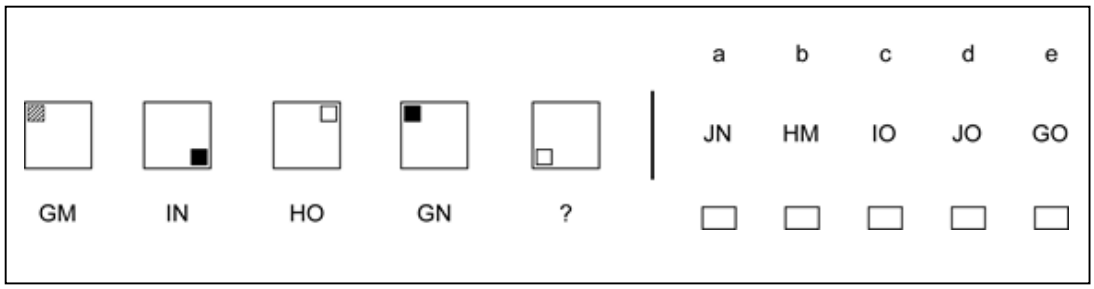
4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

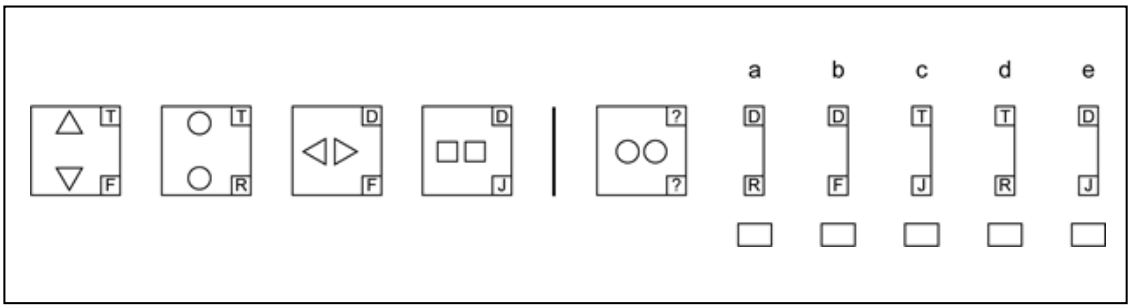
Example



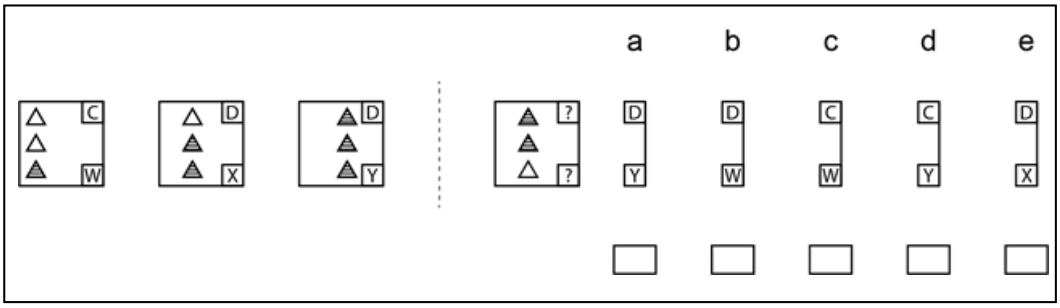
1



2



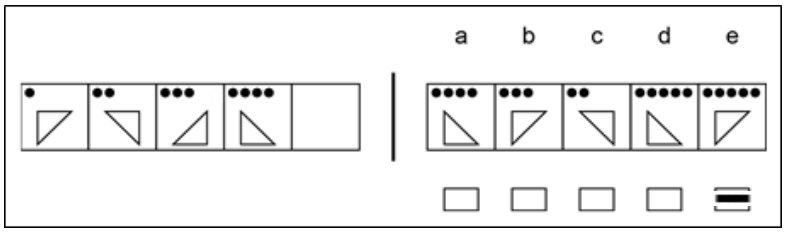
3



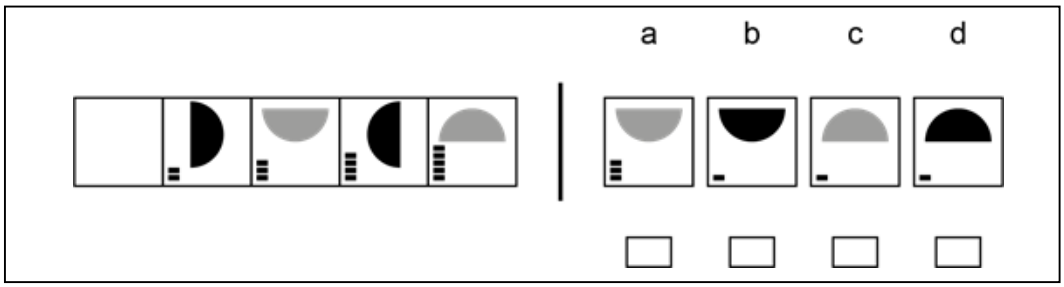
5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

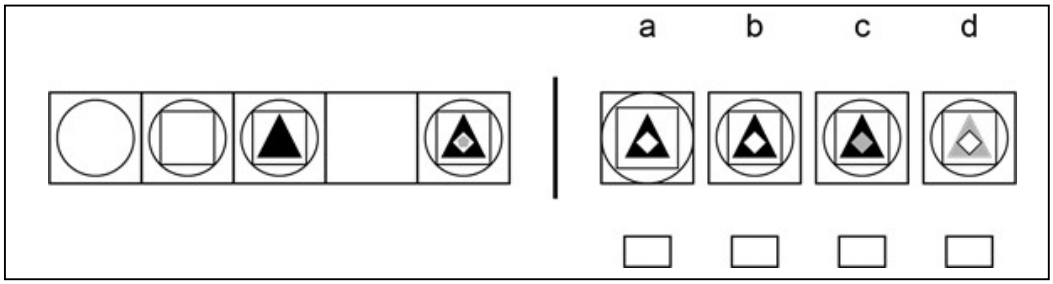
Example



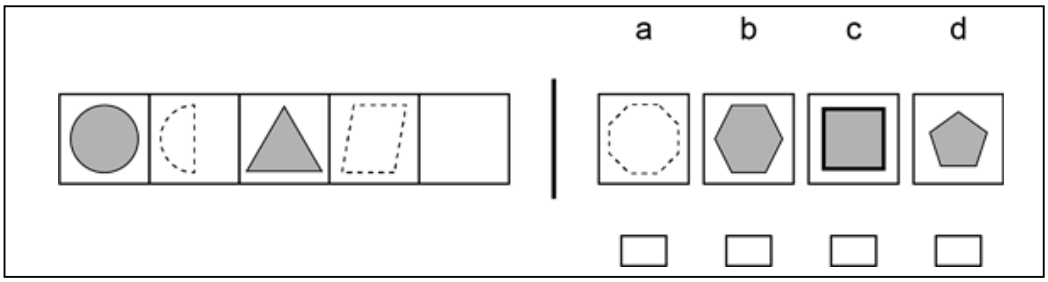
1



2



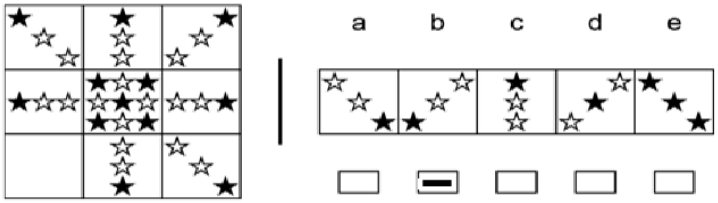
3



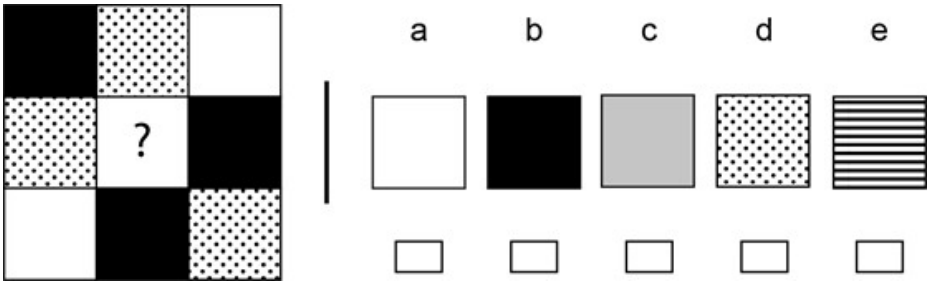
6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

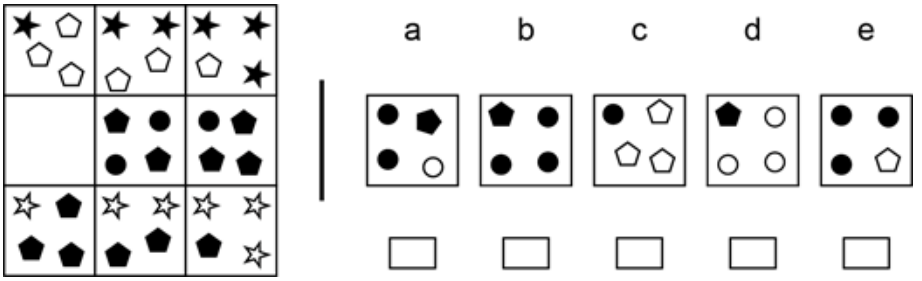
Example



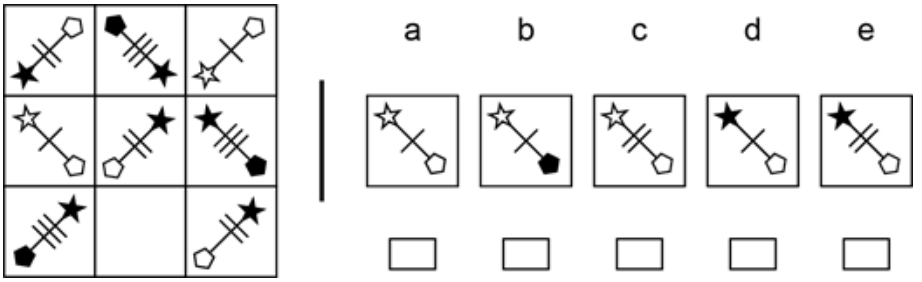
1



2



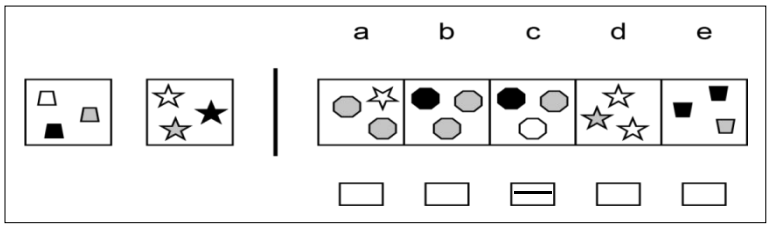
3



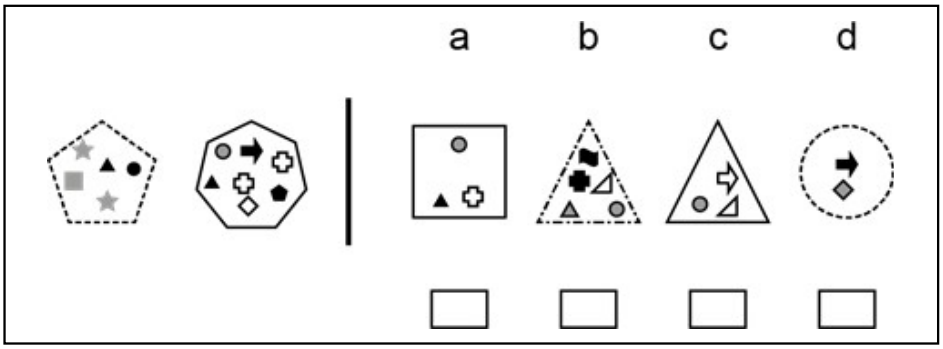
7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

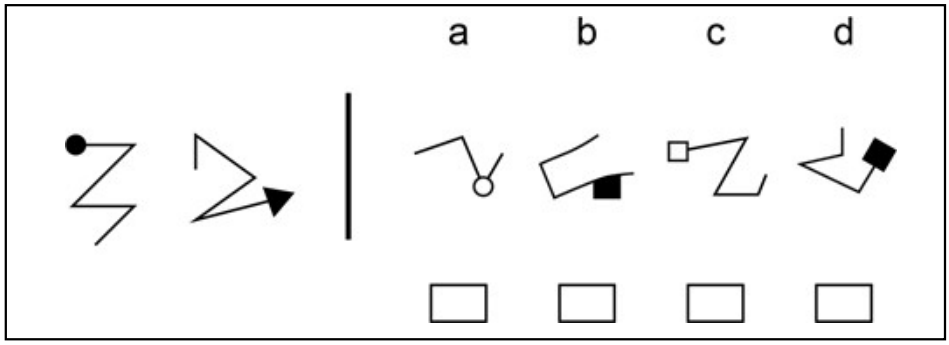
Example



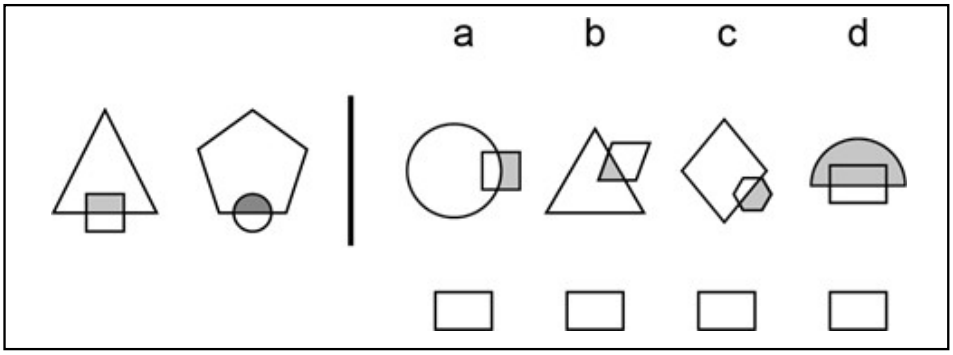
1



2



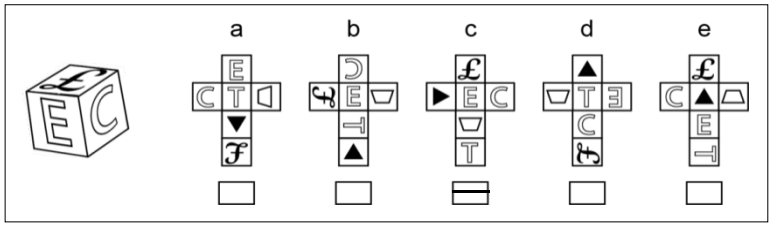
3



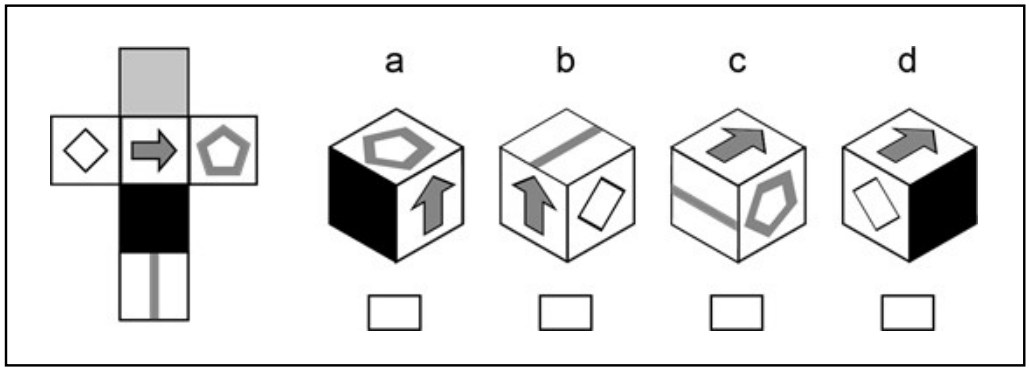
8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

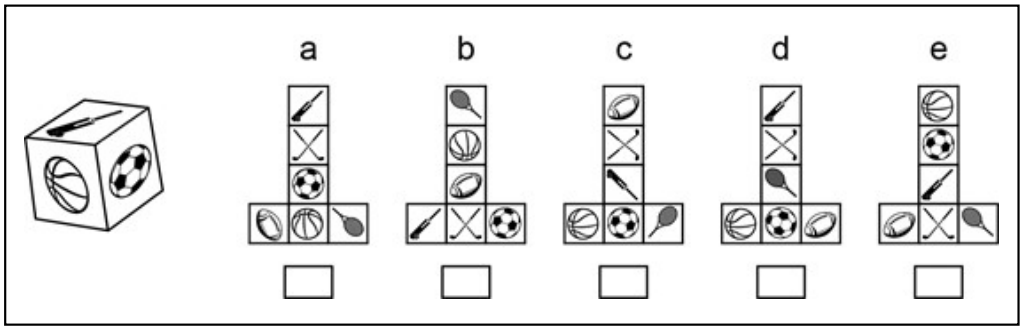
Example



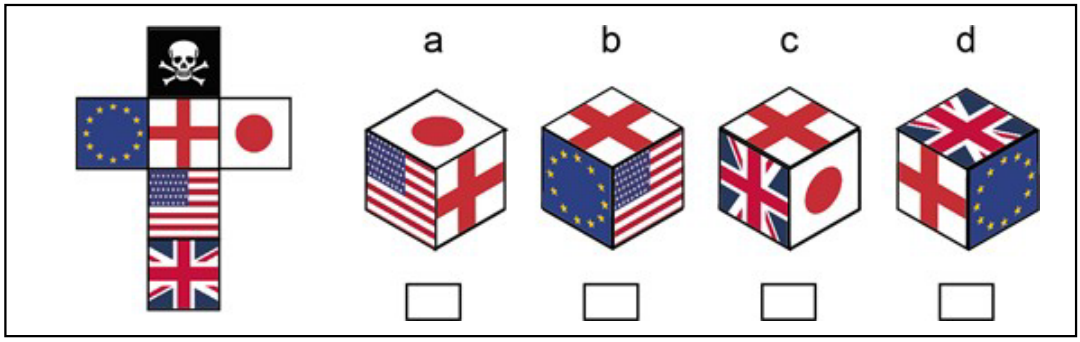
1



2



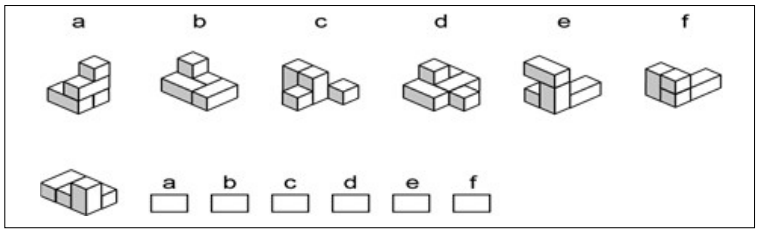
3



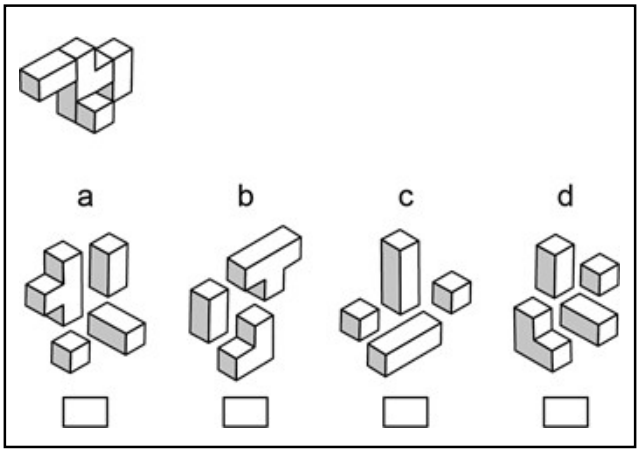
9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

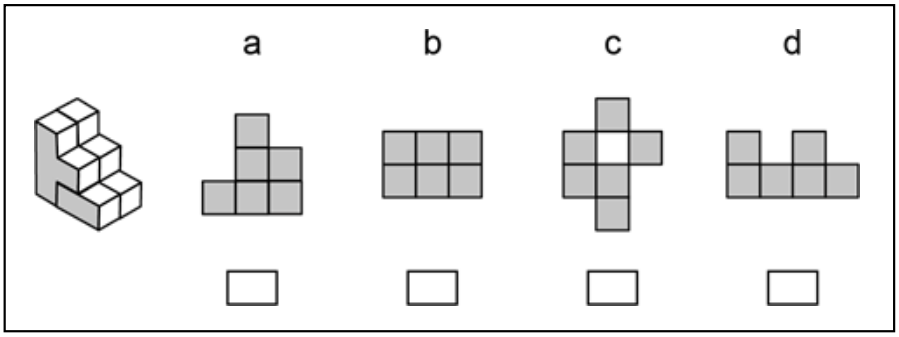
Example



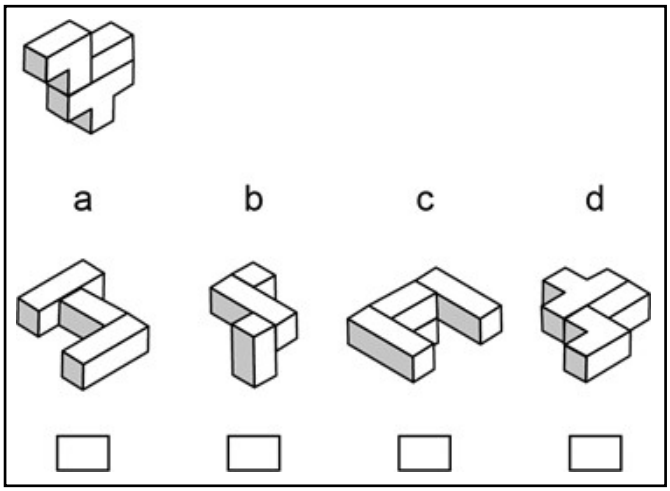
1



2



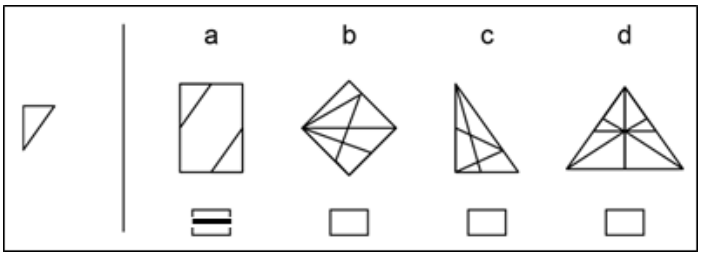
3



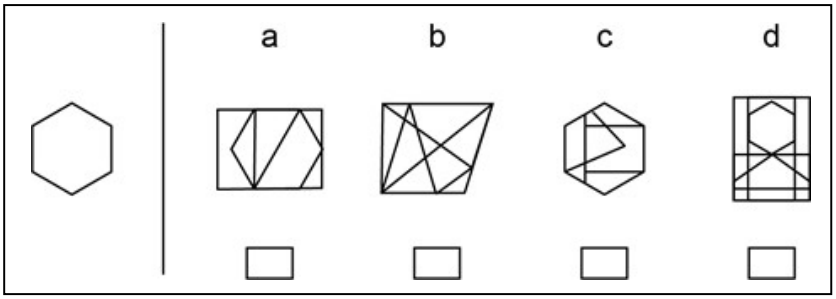
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

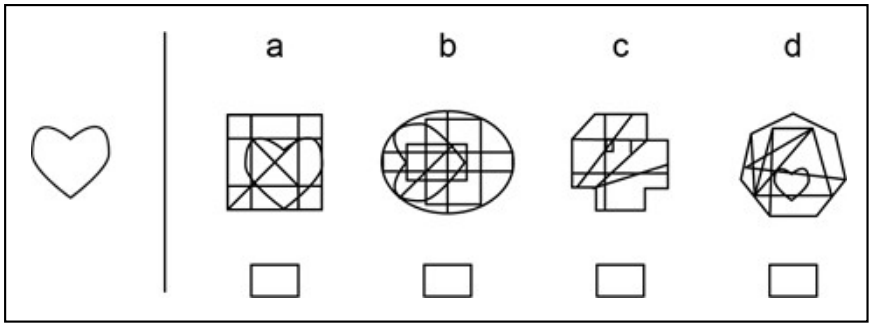
Example



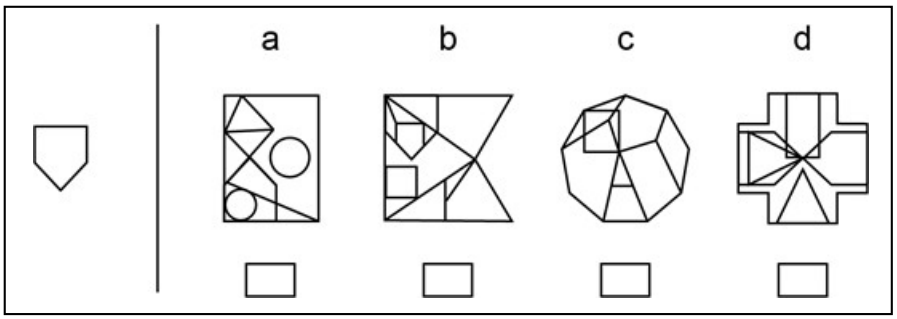
1



2



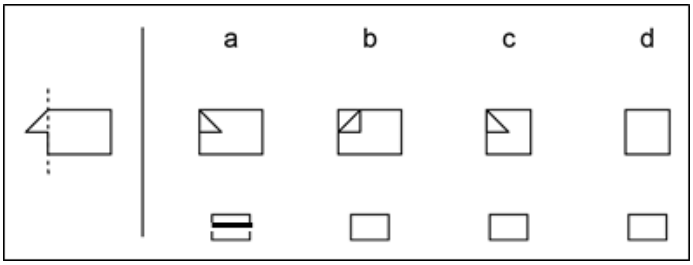
3



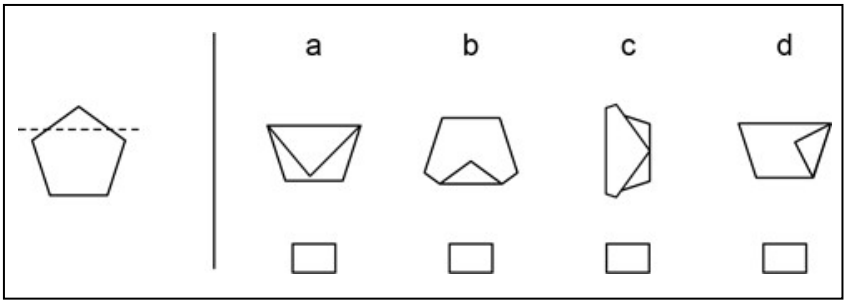
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

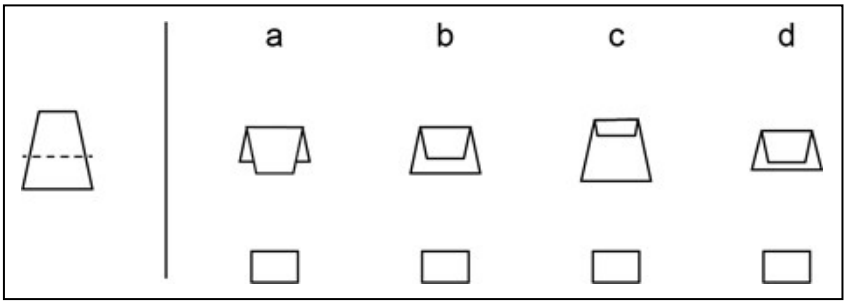
Example



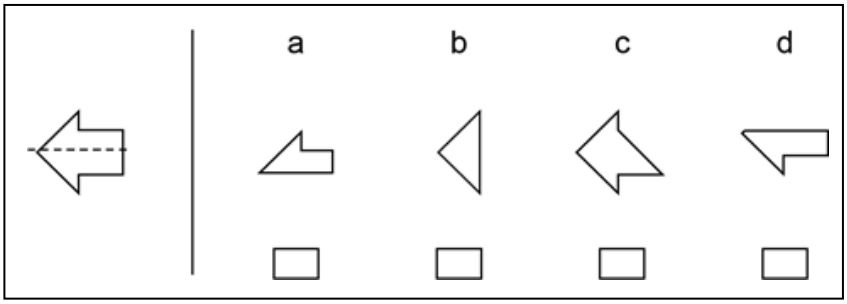
1



2



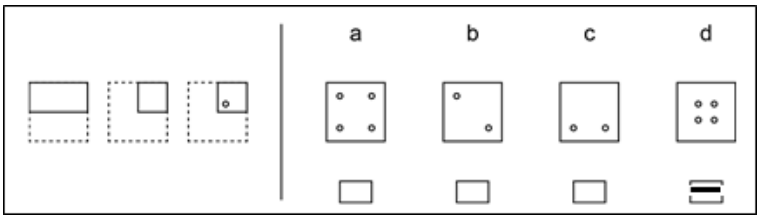
3



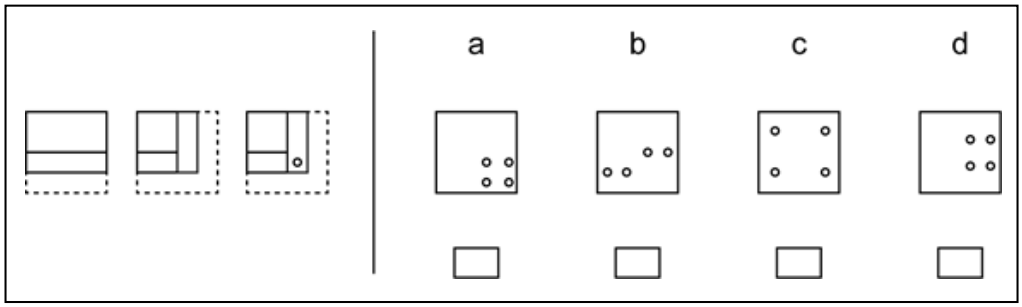
12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

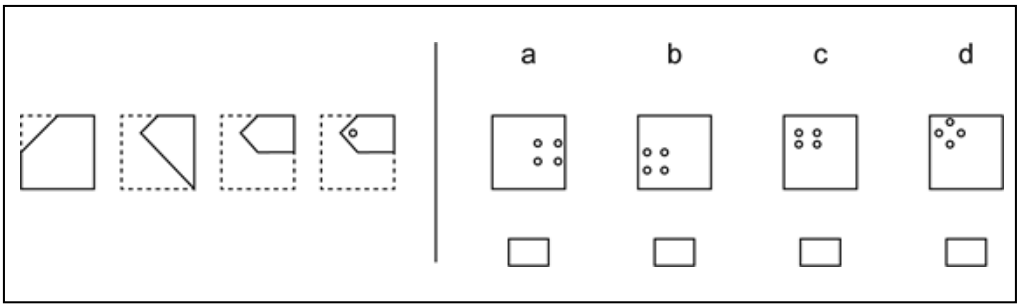
Example



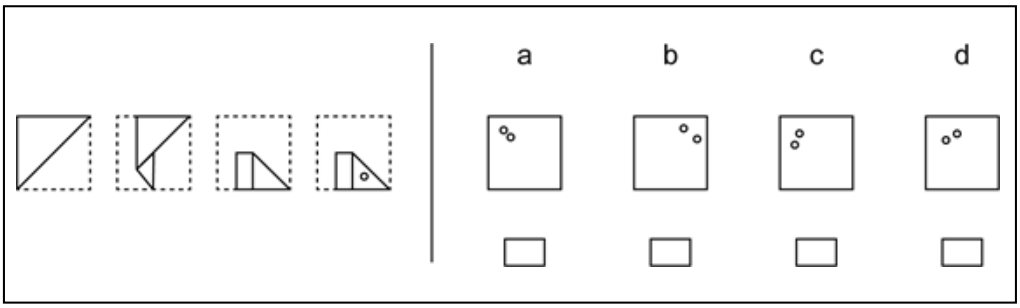
1



2



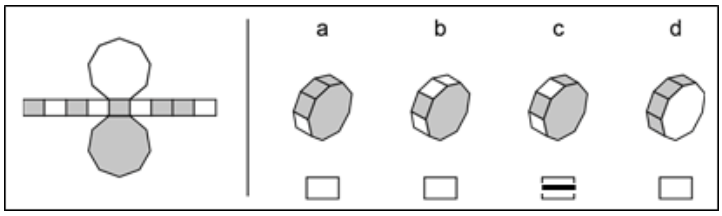
3



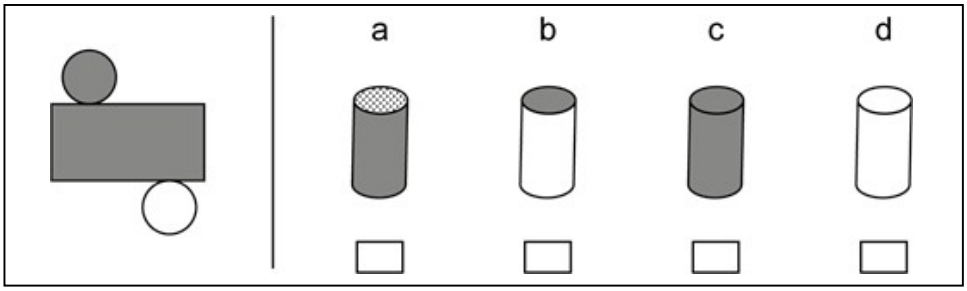
13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

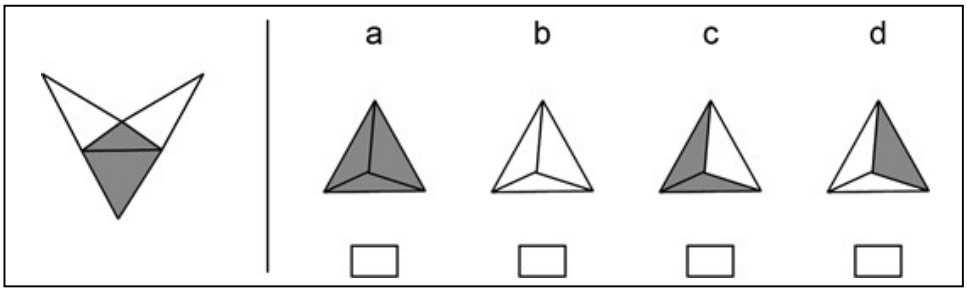
Example



1



2



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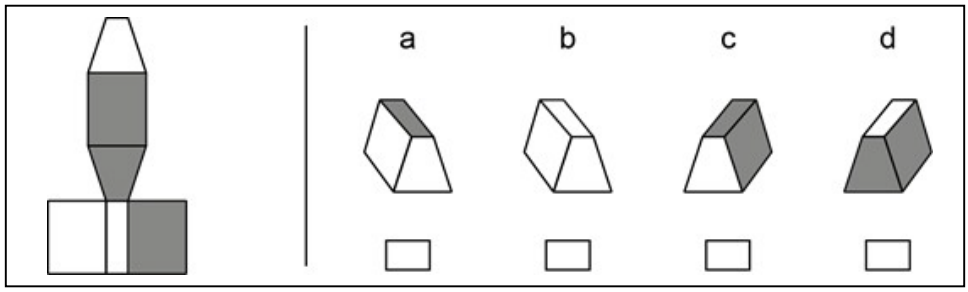


Non-Verbal Reasoning Paper 10

Like Shapes

1. **B** - (The outline of the larger shape changes from solid to dashed, while the outline of the smaller shape changes from dashed to solid. The smaller shape also moves to the opposite top corner.)
2. **B** – (Two changes occur. First, the shading of the two halves swaps around. Second, three smaller shapes become one larger shape in the same position. Therefore the correct answer must be black in the left half and grey in the right half, with one large diamond in the right half.)
3. **D** - (This is a tricky question requiring close observation of how the second shape is formed from the first shape in the example. The bottom shape moves to the top in the same orientation, while the top shape moves to the bottom, also rotating 180 degrees.)

Odd One Out

1. **B** – (All options are a rotation of a question mark apart from B, which is a reflection.)
2. **D** – (All answer options contain three shapes of the same type in different sizes apart from D, which contains one octagon and two hexagons.)
3. **B** – (In all answer options two shapes of the same type touch the circumference of the circle in the same way apart from B. The lines inside the circles are not relevant.)

Rotations

1. **E** – (E is a rotation of approximately 60 degrees clockwise.)
2. **B** – (B is a rotation of 90 degrees clockwise.)
3. **A** – (A is a rotation of 90 degrees clockwise.)

Code Breaker

1. **C** – (You can work out that K stands for the small shape being at the shape as there are two of them. Therefore S stands for the small shape being to the left of the shape. You can then work out that T stands for the small

shape being a square as there are two of them. Therefore J stands for the small shape being a triangle, making the answer SJ.)

2. **E** - (You can work out that the bottom letter represents the number of dots in the middle of the triangle as both the first and third shapes have R as their bottom letter and both have two dots in the middle. You can then work out that the top letter must represent the type of triangle as all shapes have a different top letter and the only other difference between them is that one is an equilateral triangle, one is a right angled triangle and one is a scalene triangle. Therefore the answer is BO.)
3. **B** - (You can work out that I stands for the shape of the flag being wavy as there are two of them. Therefore O stands for the shape of the flag being a rectangle. You can then work out that G stands for the pattern of the flag being the England flag as there are two of them. Therefore C stands for the pattern being the Sweden flag, making the answer OC.)

Missing Sequence

1. **B** - (As you move through the sequence four changes occur. First, the small circle alternates in colour between black and white. Second, the small circle moves round the square by one corner at each step in an anti-clockwise direction. Third the number of small lines increases by one at each step. Fourth, the star pattern alternates between one white star and one white star with one black star underneath it. Therefore the correct answer must have a small black circle in the bottom right corner, three small lines and one white star.)
2. **B** - (As you move through the sequence four changes occur. First, the star alternates in colour between grey and black. Second, the star moves its position at each step in the pattern top, middle, bottom, middle, top. Third, the diamond moves its position at each step in the pattern left, middle, right, middle, left. Fourth, the trapezium rotates by 90 degrees clockwise at each step. When all of these changes are considered, the final step in the sequence will actually be identical to the first step!)
3. **C** - (As you move through the sequence three changes occur. First, the number of sides of the outer shape decreases by one at each step. Second, the inner oval alternates between being horizontal and vertical. Third, the inner oval alternates between being grey and white. Therefore

the correct answer must be a six-sided shape with a horizontal grey oval inside.)

Missing Square

1. **C** - (In both rows the shape on the left is the same type of shape as the shape on the right, created by a border of smaller versions of the shape on the right.)
2. **E** - (The shape on the left represents a quarter of the border of the shape on the right, while the colour of the dot on the left represents the colour of the dots in the shape on the right.)
3. **B** - (In each of the corner squares the two innermost octagons are black, with the outermost octagon being white so that the overall pattern of the grid is a star.)

Most Like

1. **B** – (In both examples, each of the inner shapes has one less side than the outer shape. The number of inner shapes equals the number of sides of the outer shape.)
2. **B** – (In both examples a seven sided shape has a random smaller shape at one of the points.)
3. **C** – (Both examples are quadrilaterals, therefore B and C immediately appear the most like. B is more similar to the examples than C because it has a solid border and grey shading, whereas neither of the examples have a dashed border despite one of them being white.)

Nets

1. **B** - (The symbols used in this question initially look quite confusing, however if you focus on the relationship between the three faces in the given cube then the answer becomes much easier to spot. You know that Ψ must be to the left of Π, and that Ω must be directly above Ψ so that when Π folds around its top edge will be to the right of Ω. The only net that satisfies this is B.)

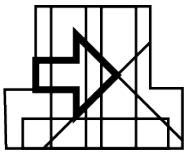
2. **D** – (You can see from the given cube that the train will look like it is travelling towards the aeroplane and the car will look like it is travelling away from the aeroplane in an opposite direction to the train. When you consider this it is much easier to spot that D is the correct answer.)
3. **C** – (You can see from the net that the arrow must be pointing towards the grey circle and that, when folded around, the black square will be to the left of the arrow.)

3D Shapes

1. **C** – (This question is tricky and it is easy to mistakenly think that A or B could be correct. You need to observe the 3D shape closely to spot that the left horizontal block will have a length of at least three cubes, and the vertical block must also have a length of at least three cubes because you can see that it is taller that the two blocks stacked on top of one another to the right. The lower block on the right will then have a length of two cubes with a single cube on top of it. Therefore C is the correct answer as there will be two longer blocks and one medium length block, rather than one longer block and two medium length blocks.)
2. **C** – (This question initially appears tricky because, from the viewpoint you are given, you cannot be sure if there are any blocks in the far corner as the view is obscured by the tallest point. However, from the view you are given you do know that if it is viewed from above in 2D there must be at least one row of three squares – you can see the tallest block with a medium height block to the left and a cube to the right. Therefore C must be correct as it is the only answer option with three squares in a row. As a quick check, you can see that the cube you can initially see furthest to the left would be represented by the square at the top left of the 2D view and you can therefore deduce that there must be an additional cube or block in the far corner that you could not initially see.)
3. **A** – (The shape has been flipped upside down and then rotated 90 degrees in the y-axis.)

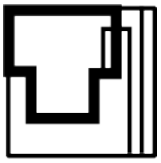
Hidden Shape

1. **A** -



2. **C** -

3. **D** –

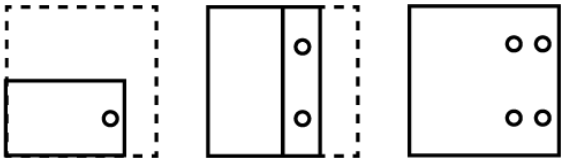


Fold Along the Dotted Line

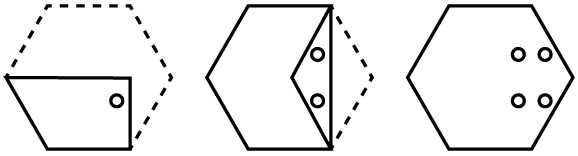
1. **B** – (The left portion of the shape will fold up over the right portion completely covering it.)
2. **C** – (The top half will fold down behind the bottom half so that the right part of the top half will be visible underneath.)
3. **A** – (The right portion will fold behind the left portion so that the top part of it will be visible underneath.)

Hole Punch

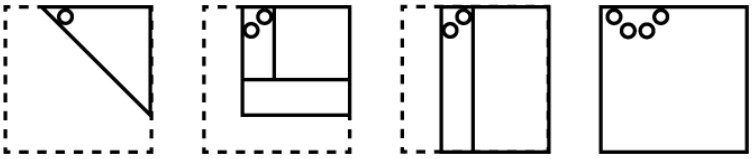
1. **D** -



2. **B** -



3. **A** –



Shaded Nets

1. **C** - (From the net you can see that the thinnest rectangular face must be green. C is the only answer option that satisfies this. You can also see from the net that when the shape is folded up so that the white five-sided face is stood up, the other two faces will be grey and green.)
2. **C** - (From the net you can see that when it is folded down and viewed from the grey side, the pattern of shading of the squares will be white, grey, grey, white, white, white, grey.)
3. **A** – (C cannot be correct as the triangle must be either black or white. You can see from the net that if the triangle is black, it must have a red face to the right and a purple face to the left. If the triangle is white it must have a red face to the left and a purple face to the right. The only answer that matches the correct orientation is A.)

Non-Verbal Reasoning Paper 11

Like Shapes

1. **A** - (A five-sided shape becomes a five-pointed star, therefore a six-sided shape becomes a six-pointed star.)
2. **B** - (The shape changes from one quarter being shaded to opposite quarters being shaded.)
3. **D** – (The shapes swap places, size and colour.)

Odd One Out

1. **E** – (In each domino the top half has double the value of the bottom half, apart from E.)
2. **E** – (The number of concentric rings corresponds to the number of sides of the shape in the middle in all apart from E, which has six concentric rings and an eight-sided shape in the middle.)
3. **D** – (In each option the stars are all the same colour apart from D, which has three white stars and one black star.)

Rotations

1. **C** – (C is a rotation of approximately 45 degrees clockwise.)
2. **B** – (This is a deliberately tricky question we have included for fun to keep you thinking on your feet. Because the shape is a wind turbine only the blades have rotated approximately 110 degrees clockwise or 10 degrees anti-clockwise, depending on which way the wind turbine is spinning! Technically the rotation could also be 230 degrees clockwise, 350 degrees clockwise, 130 degrees anti-clockwise or 250 degrees anti-clockwise due to the symmetry of the blades.)
3. **E** – (E is a rotation of 180 degrees.)

Code Breaker

1. **D** - (You can work out that G represents the small square being in the top left corner as there are two of them. As none of the four given squares have a small square in the bottom left corner, you do not know what letter will represent this, however you do know that it will be the first letter. Looking at the answer options, you know that the first letter cannot be H, I or G, therefore it must be J as that is the only other option. You can then work out that N represents the small square being black as there are two of them. Therefore O represents the small square being white, making the final answer JO.)
2. **A** - (You can work out that T stands for the shapes being apart vertically, while D stands for the shapes being together horizontally as there are two of each. You can then work out that F stands for the shapes being triangles as there are two of them. Therefore R stands for the shapes being circles, making the final answer DR.)
3. **E** – (This is a very tricky question. You can work out that D stands for the middle square being shaded as there two of them and it is the only similarity. Therefore, this means that C stands for the middle square not being shaded. All of the bottom letters are different and there are two differences between all of them. Firstly, the three squares have a different number of triangles shaded. So, W is one, X is two and Y is three. Or you could say W is the triangles being on the left hand side, X in the middle and Y on the right.

Whichever of the two methods you use, the bottom letter for the question will be X. Therefore the final answer is DX.)

Missing Sequence

1. **C** - (As you go through the sequence the number of lines in the bottom left corner increases by one each time. The semi-circle alternates in colour between grey and black and rotates by 90 degrees clockwise each time.)
2. **B** – (The sequence builds a pattern with an extra inner shape being added each time. You can see that the correct answer must start with an outer circle, middle square and inner black triangle and be able to result in an outer circle, square, black triangle, white diamond and grey pentagon. The answer cannot be A as the outer circle and square are too large.)
3. **D** – (As you move through the sequence the number of sides increases by one each time while the shading and border alternates between grey with a solid border and white with a dashed border. Therefore the correct answer must be a five-sided shape with grey shading and a solid border.)

Missing Square

1. **A** - (Each of the completed rows and each of the completed columns have one white square, one dotted square and one black square in. The missing square is in the middle row and middle column and both of these have one dotted square and one black square in so the missing square must be a white square so the answer is A.)
2. **B** - (As you move across the two completed rows from left to right, the number of one of the shapes in each square increases from 1 shape to 2 shapes to 3 shapes and the number of the other shape in each square decreases from 3 shapes to 2 shapes to 1 shape. In the middle row, the right hand square has 1 circle and 3 pentagons. Therefore, the missing square on the left should have 3 circles and 1 pentagon. The answer cannot be D as all of the shapes must be black.)
3. **A** - (As you move across the two completed rows from left to right, a number of things happen: the direction of the line alternates between left to right and right to left, the number of the small dashes on the line increases from 1

to 2 to 3 and then back to 1 etc. and the colours of the shapes on the end changes from one white and one black to both black to both white and then back to one white and one black. So the missing square should have the line pointing from the bottom right to the top left, 1 dash on the line and the shapes on the end should both be white, so the answer is A.)

Most Like

1. **C** – (The number of internal shapes equals the number of sides of the outer shape.)
2. **D** – (Both examples show four zig-zag straight lines with a solid black shape at one end.)
3. **B** – (In both examples two shapes overlap and the overlapping part is shaded.)

Nets

1. **D** - (You can see from the net that the solid black face must be underneath the arrow when it is pointing to the right, and when folded around the diamond will be to the left of the black face.)
2. **C** – (You can see from the cube that the basketball must be to the left of the football and the handle of the cricket bat must be closer to the football, while the tip of the cricket bat must be closer to the basketball. When the basketball is folded around in net C this will be achieved.)
3. **B** – (You can see from the net that the England flag must be above the USA flag and the EU flag must be to the left of the England flag. When folded around, the EU flag will also appear to the left of the USA flag. Answer A is incorrect as the USA flag is in the wrong orientation with respect to the England flag.)

3D Shapes

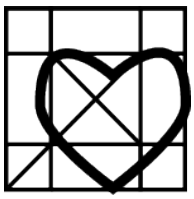
1. **A** – (Looking at the 3D shape you can see that the middle block is a T shape, with a medium sized block at either end and a single cube at the side.)

2. **B** – (This 3D shape is essentially a staircase. Therefore when it is viewed from above in 2D it will appear as a simple 2 x 3 rectangle because you will not see the effect of the 'steps'.)
3. **D** – (The shape has been pushed flat so that it is on its side.)

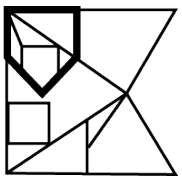
Hidden Shape

1. **C** –

2. **A** –



3. **B** –

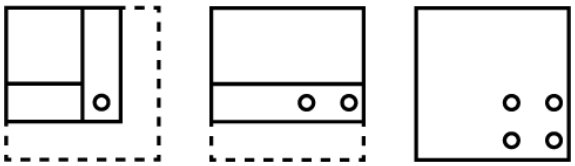


Fold Along the Dotted Line

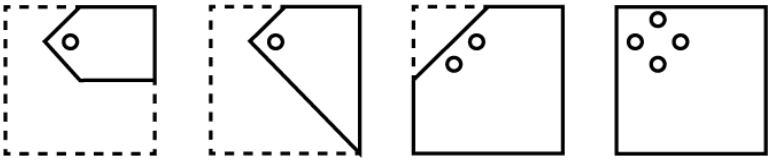
1. **B** – (The bottom portion will fold behind the top portion so that most of the bottom portion appears upside down underneath.)
2. **A** – (The top portion will fold over the bottom portion. Notice that the fold line is slightly below half way, meaning that when folded over the top portion will extend beyond the bottom of the bottom portion.)
3. **D** – (The arrow has been folded in half. A is incorrect because the arrow is not long enough.)

Hole Punch

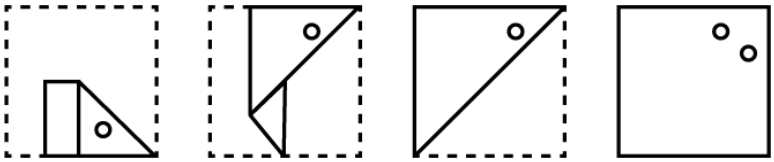
1. **A** –



2. **D** –



3. **B** –



Shaded Nets

1. **C** – (You can see from the net that the main body of the cylinder must be grey. One of the circle faces must be white and the other grey, therefore the correct answer must be C rather than A.)
2. **C** – (The small triangular base must be grey, therefore B and D are both incorrect. You can see from the net that one of the larger triangles must be grey and the other two white, therefore A cannot be correct as it has both the larger triangles grey.)
3. **B** – (From the net you know that the thin rectangle must be white, therefore A and C must be incorrect. When the net is folded down and viewed so that the thin part of the white trapezium face is at the top, the rectangular face to the left will be white.)